

Práctica Tema 5

Serafi Nebot Ginard

04/05/2025

Contents

1	Apartado 1	2
1.1	Modelo base	2
1.2	Modelo a	8
1.3	Modelo b	13
1.4	Modelo c	21
1.5	Modelo d	27
1.6	Modelo e	41
1.7	Modelo f	42

1 Apartado 1

1.1 Modelo base

La productividad del sistema es la productividad del terminal, por lo tanto: $X := \text{MTHRPUT}(\text{TERMINAL})$ (línea 24).

El tiempo de respuesta del sistema se puede calcular a partir de la ley del tiempo de respuesta interactiva: $R = \frac{N}{X} - Z$. Como X ya la hemos calculado, N es el numero total de usuarios y Z el tiempo de servicio del terminal: $R := N / X - \text{MSERVICE}(\text{TERMINAL})$ (línea 27). El tiempo de respuesta de la CPU, $R1$, y el tiempo de respuesta del disco, $R2$, se pueden obtener con la función $\text{MRESPONS}()$ (líneas 25-26).

La cantidad de usuarios en la CPU, $N1$, y la cantidad de usuarios del disco, $N2$, se pueden calcular a partir de la siguiente expresión: $N_i = X_i R_i$ (líneas 30-31).

Ejecutando el modelo se han obtenido los siguientes resultados:

N	R1	R2	R	X	N1	N2
1	0.0300	0.1000	0.9400	0.1119	0.0268	0.0783
2	0.0308	0.1078	1.0010	0.2222	0.0547	0.1677
3	0.0316	0.1168	1.0710	0.3307	0.0837	0.2703
4	0.0325	0.1270	1.1490	0.4372	0.1137	0.3888
5	0.0334	0.1389	1.2390	0.5412	0.1446	0.5261
6	0.0343	0.1526	1.3430	0.6422	0.1764	0.6860
7	0.0352	0.1686	1.4630	0.7398	0.2089	0.8731
8	0.0362	0.1873	1.6010	0.8332	0.2417	1.0920
9	0.0372	0.2092	1.7630	0.9219	0.2747	1.3500
10	0.0382	0.2350	1.9510	1.0050	0.3074	1.6530

```

1 /DECLARE/ QUEUE CPU,DISC,TERMINAL;
2     REAL Z=8.;
3     REAL V_CPU=8.,V_DISC=7.;
4     REAL X,X1,X2,R,R1,R2,N1,N2;
5     INTEGER N;
6 /STATION/ NAME=CPU;
7 &     SCHED=PS;
8     SERVICE=EXP(0.03);
9     TRANSIT=DISC,V_DISC,TERMINAL,1;
10 /STATION/ NAME=DISC;
11     TRANSIT=CPU;
12     SERVICE=EXP(0.1);
13 /STATION/ NAME=TERMINAL;
14     TYPE=INFINITE;
15     INIT=N;
16     SERVICE=EXP(Z);
17     TRANSIT=CPU;
18 /CONTROL/ CLASS=ALL QUEUE;
19 /EXEC/   FOR N:=1 STEP 1 UNTIL 10 DO
20     BEGIN
21         PRINT;
22         PRINT("NUMERO DE USUARIOS=",N);
23         SOLVE;
24
25         X := MTHRPUT(TERMINAL);
26         X1 := MTHRPUT(CPU);
27         X2 := MTHRPUT(DISC);
28         R := N / X - Z;
29         R1 := MRESPONS(CPU);
30         R2 := MRESPONS(DISC);
31         N1 := X1 * R1;

```

```

32          N2 := X2 * R2;
33
34          PRINT("R1 = ", R1);
35          PRINT("R2 = ", R2);
36          PRINT("X = ", X);
37          PRINT("R  = ", R);
38          PRINT("N1 = ", N1);
39          PRINT("N2 = ", N2);
40          END;

```

1

```

NUMERO DE USUARIOS=      1
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.2685E-01*0.2685E-01*0.3000E-01*0.8949  *
*          *          *          *          *          *
*DISC      *0.1000    *0.7830E-01*0.7830E-01*0.1000    *0.7830  *
*          *          *          *          *          *
*TERMINAL  * 8.000    *0.0000E+00*0.8949    * 8.000    *0.1119  *
*          *          *          *          *          *
*****
          MEMORY USED:      6992 WORDS OF 4 BYTES
          ( 0.14  % OF TOTAL MEMORY)

R1 =  0.3000E-01
R2 =  0.1000
X =  0.1119
R =  0.9400
N1 =  0.2685E-01
N2 =  0.7830E-01

```

1

```

NUMERO DE USUARIOS=      2
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.5333E-01*0.5476E-01*0.3081E-01* 1.778  *
*          *          *          *          *          *
*DISC      *0.1000    *0.1555    *0.1677    *0.1078    * 1.555  *
*          *          *          *          *          *
*TERMINAL  * 8.000    *0.0000E+00* 1.778    * 8.000    *0.2222  *
*          *          *          *          *          *
*****
          MEMORY USED:      7018 WORDS OF 4 BYTES
          ( 0.14  % OF TOTAL MEMORY)

R1 =  0.3081E-01
R2 =  0.1078
X =  0.2222
R =  1.001
N1 =  0.5476E-01
N2 =  0.1677

```

1

```

NUMERO DE USUARIOS=      3

```

- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*   NAME   * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*         *         *         *         *         *         *
*CPU      *0.3000E-01*0.7938E-01*0.8372E-01*0.3164E-01* 2.646 *
*         *         *         *         *         *         *
*DISC     *0.1000   *0.2315   *0.2703   *0.1168   * 2.315 *
*         *         *         *         *         *         *
*TERMINAL * 8.000   *0.0000E+00* 2.646   * 8.000   *0.3307 *
*         *         *         *         *         *         *
```

MEMORY USED: 7044 WORDS OF 4 BYTES
(0.14 % OF TOTAL MEMORY)

R1 = 0.3164E-01
R2 = 0.1168
X = 0.3307
R = 1.071
N1 = 0.8372E-01
N2 = 0.2703

1

NUMERO DE USUARIOS= 4

- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*   NAME   * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*         *         *         *         *         *         *
*CPU      *0.3000E-01*0.1049   *0.1137   *0.3251E-01* 3.498 *
*         *         *         *         *         *         *
*DISC     *0.1000   *0.3060   *0.3888   *0.1270   * 3.060 *
*         *         *         *         *         *         *
*TERMINAL * 8.000   *0.0000E+00* 3.498   * 8.000   *0.4372 *
*         *         *         *         *         *         *
```

MEMORY USED: 7070 WORDS OF 4 BYTES
(0.14 % OF TOTAL MEMORY)

R1 = 0.3251E-01
R2 = 0.1270
X = 0.4372
R = 1.149
N1 = 0.1137
N2 = 0.3888

1

NUMERO DE USUARIOS= 5

- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*   NAME   * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*         *         *         *         *         *         *
*CPU      *0.3000E-01*0.1299   *0.1446   *0.3341E-01* 4.329 *
*         *         *         *         *         *         *
*DISC     *0.1000   *0.3788   *0.5261   *0.1389   * 3.788 *
*         *         *         *         *         *         *
*TERMINAL * 8.000   *0.0000E+00* 4.329   * 8.000   *0.5412 *
*         *         *         *         *         *         *
```

MEMORY USED: 7096 WORDS OF 4 BYTES
(0.14 % OF TOTAL MEMORY)

R1 = 0.3341E-01
R2 = 0.1389
X = 0.5412
R = 1.239
N1 = 0.1446
N2 = 0.5261

1

NUMERO DE USUARIOS= 6
- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.1541  *0.1764  *0.3434E-01* 5.138 *
*          *          *          *          *          *
*DISC      *0.1000    *0.4495  *0.6860  *0.1526    * 4.495 *
*          *          *          *          *          *
*TERMINAL  * 8.000    *0.0000E+00* 5.138  * 8.000    *0.6422 *
*          *          *          *          *          *
```

MEMORY USED: 7122 WORDS OF 4 BYTES
(0.14 % OF TOTAL MEMORY)

R1 = 0.3434E-01
R2 = 0.1526
X = 0.6422
R = 1.343
N1 = 0.1764
N2 = 0.6860

1

NUMERO DE USUARIOS= 7
- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.1775  *0.2089  *0.3529E-01* 5.918 *
*          *          *          *          *          *
*DISC      *0.1000    *0.5178  *0.8731  *0.1686    * 5.178 *
*          *          *          *          *          *
*TERMINAL  * 8.000    *0.0000E+00* 5.918  * 8.000    *0.7398 *
*          *          *          *          *          *
```

MEMORY USED: 7148 WORDS OF 4 BYTES
(0.14 % OF TOTAL MEMORY)

R1 = 0.3529E-01
R2 = 0.1686
X = 0.7398
R = 1.463
N1 = 0.2089
N2 = 0.8731

1

NUMERO DE USUARIOS= 8
- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*   NAME   * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*           *         *         *         *         *         *
*CPU       *0.3000E-01*0.2000   *0.2417   *0.3627E-01* 6.666   *
*           *         *         *         *         *         *
*DISC      *0.1000   *0.5833   * 1.092   *0.1873   * 5.833   *
*           *         *         *         *         *         *
*TERMINAL  * 8.000   *0.0000E+00* 6.666   * 8.000   *0.8332   *
*           *         *         *         *         *         *
```

```
*****
MEMORY USED:      7174 WORDS OF 4 BYTES
( 0.14 % OF TOTAL MEMORY)
```

```
R1 = 0.3627E-01
R2 = 0.1873
X = 0.8332
R = 1.601
N1 = 0.2417
N2 = 1.092
```

1

```
NUMERO DE USUARIOS=      9
- MEAN VALUE ANALYSIS ("MVA") -
```

```
*****
*   NAME   * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*           *         *         *         *         *         *
*CPU       *0.3000E-01*0.2212   *0.2747   *0.3725E-01* 7.375   *
*           *         *         *         *         *         *
*DISC      *0.1000   *0.6453   * 1.350   *0.2092   * 6.453   *
*           *         *         *         *         *         *
*TERMINAL  * 8.000   *0.0000E+00* 7.375   * 8.000   *0.9219   *
*           *         *         *         *         *         *
```

```
*****
MEMORY USED:      7200 WORDS OF 4 BYTES
( 0.14 % OF TOTAL MEMORY)
```

```
R1 = 0.3725E-01
R2 = 0.2092
X = 0.9219
R = 1.763
N1 = 0.2747
N2 = 1.350
```

1

```
NUMERO DE USUARIOS=     10
- MEAN VALUE ANALYSIS ("MVA") -
```

```
*****
*   NAME   * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*           *         *         *         *         *         *
*CPU       *0.3000E-01*0.2412   *0.3074   *0.3824E-01* 8.039   *
*           *         *         *         *         *         *
*DISC      *0.1000   *0.7034   * 1.653   *0.2350   * 7.034   *
*           *         *         *         *         *         *
*TERMINAL  * 8.000   *0.0000E+00* 8.039   * 8.000   * 1.005   *
*           *         *         *         *         *         *
```

```
*****
MEMORY USED:      7226 WORDS OF 4 BYTES
```

```
      ( 0.14 % OF TOTAL MEMORY)
R1 = 0.3824E-01
R2 = 0.2350
X = 1.005
R = 1.951
N1 = 0.3074
N2 = 1.653
41 /END/
```

1.2 Modelo a

Las demandas de cada dispositivo se calculan: $D_i = V_i \cdot S_i$ (líneas 34-35).

La demanda total es la suma de todas las demandas: $D = \sum_{i=1}^K D_i$ (línea 36).

La demanda del cuello de botella es la demanda máxima de todos los dispositivos: $D_b = \max_{i=1 \dots K} \{D_i\}$ (línea 37).

El punto de saturación: $N^* = \left\lceil \frac{D+Z}{D_b} \right\rceil$ (línea 38).

```

1 /DECLARE/ QUEUE CPU,DISC,TERMINAL;
2     REAL Z=8.;
3     REAL V_CPU=8.,V_DISC=7.;
4     REAL X,X1,X2,R,R1,R2,N1,N2;
5     REAL D,D1,D2,Db,Ns;
6     INTEGER N;
7 /STATION/ NAME=CPU;
8 &     SCHED=PS;
9     SERVICE=EXP(0.03);
10    TRANSIT=DISC,V_DISC,TERMINAL,1;
11 /STATION/ NAME=DISC;
12    TRANSIT=CPU;
13    SERVICE=EXP(0.1);
14 /STATION/ NAME=TERMINAL;
15    TYPE=INFINITE;
16    INIT=N;
17    SERVICE=EXP(Z);
18    TRANSIT=CPU;
19 /CONTROL/ CLASS=ALL QUEUE;
20 /EXEC/    FOR N:=1 STEP 1 UNTIL 10 DO
21    BEGIN
22        PRINT;
23        PRINT("NUMERO DE USUARIOS=",N);
24        SOLVE;
25
26        X := MTHRUPUT(TERMINAL);
27        X1 := MTHRUPUT(CPU);
28        X2 := MTHRUPUT(DISC);
29        R := N / X - Z;
30        R1 := MRESPONS(CPU);
31        R2 := MRESPONS(DISC);
32        N1 := X1 * R1;
33        N2 := X2 * R2;
34        D1 := MSERVICE(CPU) * V_CPU;
35        D2 := MSERVICE(DISC) * V_DISC;
36        D := D1 + D2;
37        Db := MAX(D1,D2);
38        Ns := INTROUND((D + Z) / Db);
39
40        PRINT("R1 = ", R1);
41        PRINT("R2 = ", R2);
42        PRINT("X = ", X);
43        PRINT("R = ", R);
44        PRINT("N1 = ", N1);
45        PRINT("N2 = ", N2);
46        PRINT("D1 = ", D1);
47        PRINT("D2 = ", D2);
48        PRINT("D = ", D);

```



```

49          PRINT("Db = ", Db);
50          PRINT("Ns = ", Ns);
51          END;

```

1

```

NUMERO DE USUARIOS=      1
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.2685E-01*0.2685E-01*0.3000E-01*0.8949 *
*          *          *          *          *          *
*DISC      *0.1000   *0.7830E-01*0.7830E-01*0.1000   *0.7830 *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00*0.8949   * 8.000   *0.1119 *
*          *          *          *          *          *
*****
          MEMORY USED:      7158 WORDS OF 4 BYTES
          ( 0.14 % OF TOTAL MEMORY)

R1 = 0.3000E-01
R2 = 0.1000
X = 0.1119
R = 0.9400
N1 = 0.2685E-01
N2 = 0.7830E-01
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000
Ns = 13.00

```

1

```

NUMERO DE USUARIOS=      2
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.5333E-01*0.5476E-01*0.3081E-01* 1.778 *
*          *          *          *          *          *
*DISC      *0.1000   *0.1555   *0.1677   *0.1078   * 1.555 *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 1.778   * 8.000   *0.2222 *
*          *          *          *          *          *
*****
          MEMORY USED:      7184 WORDS OF 4 BYTES
          ( 0.14 % OF TOTAL MEMORY)

R1 = 0.3081E-01
R2 = 0.1078
X = 0.2222
R = 1.001
N1 = 0.5476E-01
N2 = 0.1677
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000

```

Ns = 13.00

1

NUMERO DE USUARIOS= 3

- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *         *
*CPU       *0.3000E-01*0.7938E-01*0.8372E-01*0.3164E-01* 2.646 *
*          *         *         *         *         *         *
*DISC      *0.1000   *0.2315   *0.2703   *0.1168   * 2.315 *
*          *         *         *         *         *         *
*TERMINAL  * 8.000   *0.0000E+00* 2.646   * 8.000   *0.3307 *
*          *         *         *         *         *         *
```

MEMORY USED: 7210 WORDS OF 4 BYTES
(0.14 % OF TOTAL MEMORY)

R1 = 0.3164E-01
R2 = 0.1168
X = 0.3307
R = 1.071
N1 = 0.8372E-01
N2 = 0.2703
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000
Ns = 13.00

1

NUMERO DE USUARIOS= 4

- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *         *
*CPU       *0.3000E-01*0.1049   *0.1137   *0.3251E-01* 3.498 *
*          *         *         *         *         *         *
*DISC      *0.1000   *0.3060   *0.3888   *0.1270   * 3.060 *
*          *         *         *         *         *         *
*TERMINAL  * 8.000   *0.0000E+00* 3.498   * 8.000   *0.4372 *
*          *         *         *         *         *         *
```

MEMORY USED: 7236 WORDS OF 4 BYTES
(0.14 % OF TOTAL MEMORY)

R1 = 0.3251E-01
R2 = 0.1270
X = 0.4372
R = 1.149
N1 = 0.1137
N2 = 0.3888
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000
Ns = 13.00

```

1
NUMERO DE USUARIOS=      5
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *         *         *          *          *
*CPU       *0.3000E-01*0.1299   *0.1446   *0.3341E-01*  4.329   *
*          *         *         *          *          *
*DISC      *0.1000   *0.3788   *0.5261   *0.1389   *  3.788   *
*          *         *         *          *          *
*TERMINAL  * 8.000   *0.0000E+00*  4.329   * 8.000   *0.5412   *
*          *         *         *          *          *
*****
          MEMORY USED:      7262 WORDS OF 4 BYTES
          ( 0.15 % OF TOTAL MEMORY)

R1 =  0.3341E-01
R2 =  0.1389
X =  0.5412
R =  1.239
N1 =  0.1446
N2 =  0.5261
D1 =  0.2400
D2 =  0.7000
D =  0.9400
Db =  0.7000
Ns =  13.00

```

```

1
NUMERO DE USUARIOS=      6
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *         *         *          *          *
*CPU       *0.3000E-01*0.1541   *0.1764   *0.3434E-01*  5.138   *
*          *         *         *          *          *
*DISC      *0.1000   *0.4495   *0.6860   *0.1526   *  4.495   *
*          *         *         *          *          *
*TERMINAL  * 8.000   *0.0000E+00*  5.138   * 8.000   *0.6422   *
*          *         *         *          *          *
*****
          MEMORY USED:      7288 WORDS OF 4 BYTES
          ( 0.15 % OF TOTAL MEMORY)

R1 =  0.3434E-01
R2 =  0.1526
X =  0.6422
R =  1.343
N1 =  0.1764
N2 =  0.6860
D1 =  0.2400
D2 =  0.7000
D =  0.9400
Db =  0.7000
Ns =  13.00

```

```

1
NUMERO DE USUARIOS=      7

```

- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*   NAME   * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*           *         *         *         *         *         *
*CPU       *0.3000E-01*0.1775   *0.2089   *0.3529E-01* 5.918   *
*           *         *         *         *         *         *
*DISC      *0.1000   *0.5178   *0.8731   *0.1686   * 5.178   *
*           *         *         *         *         *         *
*TERMINAL  * 8.000   *0.0000E+00* 5.918   * 8.000   *0.7398   *
*           *         *         *         *         *         *
```

MEMORY USED: 7314 WORDS OF 4 BYTES
(0.15 % OF TOTAL MEMORY)

R1 = 0.3529E-01
R2 = 0.1686
X = 0.7398
R = 1.463
N1 = 0.2089
N2 = 0.8731
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000
Ns = 13.00

1

NUMERO DE USUARIOS= 8

- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*   NAME   * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*           *         *         *         *         *         *
*CPU       *0.3000E-01*0.2000   *0.2417   *0.3627E-01* 6.666   *
*           *         *         *         *         *         *
*DISC      *0.1000   *0.5833   * 1.092   *0.1873   * 5.833   *
*           *         *         *         *         *         *
*TERMINAL  * 8.000   *0.0000E+00* 6.666   * 8.000   *0.8332   *
*           *         *         *         *         *         *
```

MEMORY USED: 7340 WORDS OF 4 BYTES
(0.15 % OF TOTAL MEMORY)

R1 = 0.3627E-01
R2 = 0.1873
X = 0.8332
R = 1.601
N1 = 0.2417
N2 = 1.092
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000
Ns = 13.00

1

NUMERO DE USUARIOS= 9

- MEAN VALUE ANALYSIS ("MVA") -

```
*****
```

```

*   NAME   *   SERVICE * BUSY PCT *   CUST NB * RESPONSE *   THRUPUT *
*****
*           *           *           *           *           *           *
*CPU       *0.3000E-01*0.2212   *0.2747   *0.3725E-01* 7.375   *
*           *           *           *           *           *           *
*DISC      *0.1000   *0.6453   * 1.350   *0.2092   * 6.453   *
*           *           *           *           *           *           *
*TERMINAL  * 8.000   *0.0000E+00* 7.375   * 8.000   *0.9219   *
*           *           *           *           *           *           *
*****
                MEMORY USED:      7366 WORDS OF 4 BYTES
                ( 0.15 % OF TOTAL MEMORY)

R1 = 0.3725E-01
R2 = 0.2092
X = 0.9219
R = 1.763
N1 = 0.2747
N2 = 1.350
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000
Ns = 13.00

```

1

```

NUMERO DE USUARIOS=      10
- MEAN VALUE ANALYSIS ("MVA") -
*****
*   NAME   *   SERVICE * BUSY PCT *   CUST NB * RESPONSE *   THRUPUT *
*****
*           *           *           *           *           *           *
*CPU       *0.3000E-01*0.2412   *0.3074   *0.3824E-01* 8.039   *
*           *           *           *           *           *           *
*DISC      *0.1000   *0.7034   * 1.653   *0.2350   * 7.034   *
*           *           *           *           *           *           *
*TERMINAL  * 8.000   *0.0000E+00* 8.039   * 8.000   * 1.005   *
*           *           *           *           *           *           *
*****
                MEMORY USED:      7392 WORDS OF 4 BYTES
                ( 0.15 % OF TOTAL MEMORY)

R1 = 0.3824E-01
R2 = 0.2350
X = 1.005
R = 1.951
N1 = 0.3074
N2 = 1.653
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000
Ns = 13.00
52 /END/

```

1.3 Modelo b

El tiempo de respuesta del sistema, R , ya lo tenemos calculado a partir del modelo base (línea 30).

El tiempo de respuesta total, $R_t = R + Z$, se calcula en la línea 40.

Para la cantidad de usuarios que están trabajando, N_w , y están pensando, N_z , podemos utilizar la ley de Little: $N_i = X_i \cdot R_i$ (líneas 41-42).

Ejecutando el modelo obtenemos los siguientes resultados:

R	Rt	Nw	Nz
0.9400	8.9400	0.1051	0.8949
1.0010	9.0010	0.2225	1.778
1.0710	9.0710	0.3541	2.646
1.1490	9.1490	0.5025	3.498
1.2390	9.2390	0.6707	4.329
1.3430	9.3430	0.8624	5.138
1.4630	9.4630	1.0820	5.918
1.6010	9.6010	1.3340	6.666
1.7630	9.7630	1.6250	7.375
1.9510	9.9510	1.9610	8.039

```

1 /DECLARE/ QUEUE CPU,DISC,TERMINAL;
2     REAL Z=8.;
3     REAL V_CPU=8.,V_DISC=7.;
4     REAL X,X1,X2,R,R1,R2,N1,N2;
5     REAL D,D1,D2,Db,Ns;
6     REAL Rt,Nw,Nz;
7     INTEGER N;
8 /STATION/ NAME=CPU;
9 &        SCHED=PS;
10        SERVICE=EXP(0.03);
11        TRANSIT=DISC,V_DISC,TERMINAL,1;
12 /STATION/ NAME=DISC;
13        TRANSIT=CPU;
14        SERVICE=EXP(0.1);
15 /STATION/ NAME=TERMINAL;
16        TYPE=INFINITE;
17        INIT=N;
18        SERVICE=EXP(Z);
19        TRANSIT=CPU;
20 /CONTROL/ CLASS=ALL QUEUE;
21 /EXEC/   FOR N:=1 STEP 1 UNTIL 10 DO
22         BEGIN
23             PRINT;
24             PRINT("NUMERO DE USUARIOS=",N);
25             SOLVE;
26
27             X := MTHRUPUT(TERMINAL);
28             X1 := MTHRUPUT(CPU);
29             X2 := MTHRUPUT(DISC);
30             R := N / X - Z;
31             R1 := MRESPONS(CPU);
32             R2 := MRESPONS(DISC);
33             N1 := X1 * R1;
34             N2 := X2 * R2;
35             D1 := MSERVICE(CPU) * V_CPU;
36             D2 := MSERVICE(DISC) * V_DISC;
37             D := D1 + D2;
38             Db := MAX(D1,D2);
39             Ns := INTROUND((D + Z) / Db);
40             Rt := R + Z;

```

```

41      Nw := X * R;
42      Nz := X * Z;
43
44      PRINT("R1 = ", R1);
45      PRINT("R2 = ", R2);
46      PRINT("X = ", X);
47      PRINT("R = ", R);
48      PRINT("N1 = ", N1);
49      PRINT("N2 = ", N2);
50      PRINT("D1 = ", D1);
51      PRINT("D2 = ", D2);
52      PRINT("D = ", D);
53      PRINT("Db = ", Db);
54      PRINT("Ns = ", Ns);
55      PRINT("Rt = ", Rt);
56      PRINT("Nw = ", Nw);
57      PRINT("Nz = ", Nz);
58      END;

```

1

```

NUMERO DE USUARIOS=      1
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.2685E-01*0.2685E-01*0.3000E-01*0.8949 *
*          *          *          *          *          *
*DISC      *0.1000   *0.7830E-01*0.7830E-01*0.1000   *0.7830 *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00*0.8949   * 8.000   *0.1119 *
*          *          *          *          *          *
*****
      MEMORY USED:      7248 WORDS OF 4 BYTES
      ( 0.14 % OF TOTAL MEMORY)

R1 =  0.3000E-01
R2 =  0.1000
X =  0.1119
R =  0.9400
N1 =  0.2685E-01
N2 =  0.7830E-01
D1 =  0.2400
D2 =  0.7000
D =  0.9400
Db =  0.7000
Ns =  13.00
Rt =  8.940
Nw =  0.1051
Nz =  0.8949

```

1

```

NUMERO DE USUARIOS=      2
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.5333E-01*0.5476E-01*0.3081E-01* 1.778 *

```

```

*          *          *          *          *          *          *
*DISC      *0.1000    *0.1555    *0.1677    *0.1078    * 1.555    *
*          *          *          *          *          *          *
*TERMINAL  * 8.000    *0.0000E+00* 1.778    * 8.000    *0.2222    *
*          *          *          *          *          *          *
*****

```

```

      MEMORY USED:      7274 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

```

```

R1 = 0.3081E-01
R2 = 0.1078
X = 0.2222
R = 1.001
N1 = 0.5476E-01
N2 = 0.1677
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000
Ns = 13.00
Rt = 9.001
Nw = 0.2225
Nz = 1.778

```

1

```

NUMERO DE USUARIOS=      3
- MEAN VALUE ANALYSIS ("MVA") -

```

```

*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *          *
*CPU       *0.3000E-01*0.7938E-01*0.8372E-01*0.3164E-01* 2.646    *
*          *          *          *          *          *          *
*DISC      *0.1000    *0.2315    *0.2703    *0.1168    * 2.315    *
*          *          *          *          *          *          *
*TERMINAL  * 8.000    *0.0000E+00* 2.646    * 8.000    *0.3307    *
*          *          *          *          *          *          *
*****

```

```

      MEMORY USED:      7300 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

```

```

R1 = 0.3164E-01
R2 = 0.1168
X = 0.3307
R = 1.071
N1 = 0.8372E-01
N2 = 0.2703
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000
Ns = 13.00
Rt = 9.071
Nw = 0.3541
Nz = 2.646

```

1

```

NUMERO DE USUARIOS=      4
- MEAN VALUE ANALYSIS ("MVA") -

```

```

*****

```



```

*   NAME   *   SERVICE * BUSY PCT *   CUST NB * RESPONSE *   THRUPUT *
*****
*           *           *           *           *           *           *
*CPU       *0.3000E-01*0.1049   *0.1137   *0.3251E-01* 3.498   *
*           *           *           *           *           *           *
*DISC      *0.1000   *0.3060   *0.3888   *0.1270   * 3.060   *
*           *           *           *           *           *           *
*TERMINAL  * 8.000   *0.0000E+00* 3.498   * 8.000   *0.4372   *
*           *           *           *           *           *           *
*****

```

MEMORY USED: 7326 WORDS OF 4 BYTES
(0.15 % OF TOTAL MEMORY)

```

R1 = 0.3251E-01
R2 = 0.1270
X = 0.4372
R = 1.149
N1 = 0.1137
N2 = 0.3888
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000
Ns = 13.00
Rt = 9.149
Nw = 0.5025
Nz = 3.498

```

1

NUMERO DE USUARIOS= 5
- MEAN VALUE ANALYSIS ("MVA") -

```

*****
*   NAME   *   SERVICE * BUSY PCT *   CUST NB * RESPONSE *   THRUPUT *
*****
*           *           *           *           *           *           *
*CPU       *0.3000E-01*0.1299   *0.1446   *0.3341E-01* 4.329   *
*           *           *           *           *           *           *
*DISC      *0.1000   *0.3788   *0.5261   *0.1389   * 3.788   *
*           *           *           *           *           *           *
*TERMINAL  * 8.000   *0.0000E+00* 4.329   * 8.000   *0.5412   *
*           *           *           *           *           *           *
*****

```

MEMORY USED: 7352 WORDS OF 4 BYTES
(0.15 % OF TOTAL MEMORY)

```

R1 = 0.3341E-01
R2 = 0.1389
X = 0.5412
R = 1.239
N1 = 0.1446
N2 = 0.5261
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000
Ns = 13.00
Rt = 9.239
Nw = 0.6707
Nz = 4.329

```

1

```

NUMERO DE USUARIOS=      6
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.1541   *0.1764   *0.3434E-01*  5.138   *
*          *          *          *          *          *
*DISC      *0.1000   *0.4495   *0.6860   *0.1526   *  4.495   *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00*  5.138   *  8.000   *0.6422   *
*          *          *          *          *          *
*****
          MEMORY USED:      7378 WORDS OF 4 BYTES
          ( 0.15 % OF TOTAL MEMORY)

R1 =  0.3434E-01
R2 =  0.1526
X =  0.6422
R =  1.343
N1 =  0.1764
N2 =  0.6860
D1 =  0.2400
D2 =  0.7000
D =  0.9400
Db =  0.7000
Ns =  13.00
Rt =  9.343
Nw =  0.8624
Nz =  5.138

```

1

```

NUMERO DE USUARIOS=      7
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.1775   *0.2089   *0.3529E-01*  5.918   *
*          *          *          *          *          *
*DISC      *0.1000   *0.5178   *0.8731   *0.1686   *  5.178   *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00*  5.918   *  8.000   *0.7398   *
*          *          *          *          *          *
*****
          MEMORY USED:      7404 WORDS OF 4 BYTES
          ( 0.15 % OF TOTAL MEMORY)

R1 =  0.3529E-01
R2 =  0.1686
X =  0.7398
R =  1.463
N1 =  0.2089
N2 =  0.8731
D1 =  0.2400
D2 =  0.7000
D =  0.9400
Db =  0.7000
Ns =  13.00

```

Rt = 9.463
 Nw = 1.082
 Nz = 5.918

1

```

NUMERO DE USUARIOS=      8
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.2000   *0.2417   *0.3627E-01* 6.666   *
*          *          *          *          *          *
*DISC      *0.1000   *0.5833   * 1.092   *0.1873   * 5.833   *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 6.666   * 8.000   *0.8332   *
*          *          *          *          *          *
*****
      MEMORY USED:      7430 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

```

R1 = 0.3627E-01
 R2 = 0.1873
 X = 0.8332
 R = 1.601
 N1 = 0.2417
 N2 = 1.092
 D1 = 0.2400
 D2 = 0.7000
 D = 0.9400
 Db = 0.7000
 Ns = 13.00
 Rt = 9.601
 Nw = 1.334
 Nz = 6.666

1

```

NUMERO DE USUARIOS=      9
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.2212   *0.2747   *0.3725E-01* 7.375   *
*          *          *          *          *          *
*DISC      *0.1000   *0.6453   * 1.350   *0.2092   * 6.453   *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 7.375   * 8.000   *0.9219   *
*          *          *          *          *          *
*****
      MEMORY USED:      7456 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

```

R1 = 0.3725E-01
 R2 = 0.2092
 X = 0.9219
 R = 1.763
 N1 = 0.2747
 N2 = 1.350
 D1 = 0.2400

D2 = 0.7000
 D = 0.9400
 Db = 0.7000
 Ns = 13.00
 Rt = 9.763
 Nw = 1.625
 Nz = 7.375

1

```

NUMERO DE USUARIOS=      10
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.2412   *0.3074   *0.3824E-01* 8.039   *
*          *          *          *          *          *
*DISC      *0.1000   *0.7034   * 1.653   *0.2350   * 7.034   *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 8.039   * 8.000   * 1.005   *
*          *          *          *          *          *
*****
          MEMORY USED:      7482 WORDS OF 4 BYTES
          ( 0.15 % OF TOTAL MEMORY)

R1 = 0.3824E-01
R2 = 0.2350
X = 1.005
R = 1.951
N1 = 0.3074
N2 = 1.653
D1 = 0.2400
D2 = 0.7000
D = 0.9400
Db = 0.7000
Ns = 13.00
Rt = 9.951
Nw = 1.961
Nz = 8.039
59 /END/
  
```

1.4 Modelo c

N	R1	R2	R3	R	X	N1	N2	N3
1	0.0300	0.1000	0.1000	0.1119	0.9400	0.0268	0.0391	0.0391
2	0.0308	0.1039	0.1039	0.2229	0.9738	0.0549	0.0810	0.0810
3	0.0316	0.1081	0.1081	0.3330	1.0100	0.0843	0.1260	0.1260
4	0.0325	0.1126	0.1126	0.4421	1.0480	0.1150	0.1742	0.1742
5	0.0334	0.1174	0.1174	0.5501	1.0900	0.1472	0.2261	0.2261
6	0.0344	0.1226	0.1226	0.6569	1.1340	0.1809	0.2819	0.2819
7	0.0354	0.1282	0.1282	0.7625	1.1810	0.2161	0.3421	0.3421
8	0.0364	0.1342	0.1342	0.8666	1.2310	0.2529	0.4071	0.4071
9	0.0375	0.1407	0.1407	0.9692	1.2860	0.2915	0.4773	0.4773
10	0.0387	0.1477	0.1477	1.0700	1.3440	0.3317	0.5534	0.5534

```

1 /DECLARE/ QUEUE CPU,DISC(2),TERMINAL;
2     REAL Z=8.;
3     REAL V_CPU=8.,V_DISC(2)=(3.5,3.5);
4     REAL X,X1,X2,X3,R,R1,R2,R3,N1,N2,N3;
5     INTEGER N;
6 /STATION/ NAME=CPU;
7 &     SCHED=PS;
8     SERVICE=EXP(0.03);
9     TRANSIT=DISC,V_DISC,TERMINAL,1;
10 /STATION/ NAME=DISC;
11     TRANSIT=CPU;
12 /STATION/ NAME=DISC(1);
13     SERVICE=EXP(0.1);
14 /STATION/ NAME=DISC(2);
15     SERVICE=EXP(0.1);
16 /STATION/ NAME=TERMINAL;
17     TYPE=INFINITE;
18     INIT=N;
19     SERVICE=EXP(Z);
20     TRANSIT=CPU;
21 /CONTROL/ CLASS=ALL QUEUE;
22 /EXEC/   FOR N:=1 STEP 1 UNTIL 10 DO
23     BEGIN
24         PRINT;
25         PRINT("NUMERO DE USUARIOS=",N);
26         SOLVE;
27
28         X := MTHRUPUT(TERMINAL);
29         X1 := MTHRUPUT(CPU);
30         X2 := MTHRUPUT(DISC(1));
31         X3 := MTHRUPUT(DISC(2));
32         R := N / X - Z;
33         R1 := MRESPONS(CPU);
34         R2 := MRESPONS(DISC(1));
35         R3 := MRESPONS(DISC(2));
36         N1 := X1 * R1;
37         N2 := X2 * R2;
38         N3 := X2 * R3;
39
40         PRINT("R1 = ", R1);
41         PRINT("R2 = ", R2);
42         PRINT("R3 = ", R3);
43         PRINT("X = ", X);

```

```

44      PRINT("R = ", R);
45      PRINT("N1 = ", N1);
46      PRINT("N2 = ", N2);
47      PRINT("N3 = ", N3);
48      END;

```

1

```

NUMERO DE USUARIOS=      1
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *         *         *          *          *
*CPU       *0.3000E-01*0.2685E-01*0.2685E-01*0.3000E-01*0.8949  *
*          *         *         *          *          *
*DISC  1   *0.1000   *0.3915E-01*0.3915E-01*0.1000   *0.3915  *
*          *         *         *          *          *
*DISC  2   *0.1000   *0.3915E-01*0.3915E-01*0.1000   *0.3915  *
*          *         *         *          *          *
*TERMINAL * 8.000    *0.0000E+00*0.8949   * 8.000    *0.1119  *
*          *         *         *          *          *
*****
      MEMORY USED:      7394 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

R1 =  0.3000E-01
R2 =  0.1000
R3 =  0.1000
X =  0.1119
R =  0.9400
N1 =  0.2685E-01
N2 =  0.3915E-01
N3 =  0.3915E-01

```

1

```

NUMERO DE USUARIOS=      2
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *         *         *          *          *
*CPU       *0.3000E-01*0.5349E-01*0.5492E-01*0.3081E-01* 1.783  *
*          *         *         *          *          *
*DISC  1   *0.1000   *0.7800E-01*0.8106E-01*0.1039   *0.7800  *
*          *         *         *          *          *
*DISC  2   *0.1000   *0.7800E-01*0.8106E-01*0.1039   *0.7800  *
*          *         *         *          *          *
*TERMINAL * 8.000    *0.0000E+00* 1.783   * 8.000    *0.2229  *
*          *         *         *          *          *
*****
      MEMORY USED:      7417 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

R1 =  0.3081E-01
R2 =  0.1039
R3 =  0.1039
X =  0.2229
R =  0.9738
N1 =  0.5492E-01
N2 =  0.8106E-01

```

N3 = 0.8106E-01

1

NUMERO DE USUARIOS= 3
- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *         *
*CPU       *0.3000E-01*0.7991E-01*0.8430E-01*0.3165E-01* 2.664 *
*          *         *         *         *         *         *
*DISC  1   *0.1000   *0.1165   *0.1260   *0.1081   * 1.165 *
*          *         *         *         *         *         *
*DISC  2   *0.1000   *0.1165   *0.1260   *0.1081   * 1.165 *
*          *         *         *         *         *         *
*TERMINAL * 8.000    *0.0000E+00* 2.664   * 8.000    *0.3330 *
*          *         *         *         *         *         *
*****
```

MEMORY USED: 7440 WORDS OF 4 BYTES
(0.15 % OF TOTAL MEMORY)

R1 = 0.3165E-01
R2 = 0.1081
R3 = 0.1081
X = 0.3330
R = 1.010
N1 = 0.8430E-01
N2 = 0.1260
N3 = 0.1260

1

NUMERO DE USUARIOS= 4
- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *         *
*CPU       *0.3000E-01*0.1061   *0.1150   *0.3253E-01* 3.537 *
*          *         *         *         *         *         *
*DISC  1   *0.1000   *0.1547   *0.1742   *0.1126   * 1.547 *
*          *         *         *         *         *         *
*DISC  2   *0.1000   *0.1547   *0.1742   *0.1126   * 1.547 *
*          *         *         *         *         *         *
*TERMINAL * 8.000    *0.0000E+00* 3.537   * 8.000    *0.4421 *
*          *         *         *         *         *         *
*****
```

MEMORY USED: 7463 WORDS OF 4 BYTES
(0.15 % OF TOTAL MEMORY)

R1 = 0.3253E-01
R2 = 0.1126
R3 = 0.1126
X = 0.4421
R = 1.048
N1 = 0.1150
N2 = 0.1742
N3 = 0.1742

1

NUMERO DE USUARIOS= 5

- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*   NAME   * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*         *         *         *         *         *         *
*CPU       *0.3000E-01*0.1320   *0.1472   *0.3345E-01* 4.401   *
*         *         *         *         *         *         *
*DISC  1   *0.1000   *0.1925   *0.2261   *0.1174   * 1.925   *
*         *         *         *         *         *         *
*DISC  2   *0.1000   *0.1925   *0.2261   *0.1174   * 1.925   *
*         *         *         *         *         *         *
*TERMINAL * 8.000    *0.0000E+00* 4.401    * 8.000    *0.5501   *
*         *         *         *         *         *         *
```

MEMORY USED: 7486 WORDS OF 4 BYTES
(0.15 % OF TOTAL MEMORY)

R1 = 0.3345E-01
R2 = 0.1174
R3 = 0.1174
X = 0.5501
R = 1.090
N1 = 0.1472
N2 = 0.2261
N3 = 0.2261

1

NUMERO DE USUARIOS= 6

- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*   NAME   * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*         *         *         *         *         *         *
*CPU       *0.3000E-01*0.1577   *0.1809   *0.3442E-01* 5.255   *
*         *         *         *         *         *         *
*DISC  1   *0.1000   *0.2299   *0.2819   *0.1226   * 2.299   *
*         *         *         *         *         *         *
*DISC  2   *0.1000   *0.2299   *0.2819   *0.1226   * 2.299   *
*         *         *         *         *         *         *
*TERMINAL * 8.000    *0.0000E+00* 5.255    * 8.000    *0.6569   *
*         *         *         *         *         *         *
```

MEMORY USED: 7509 WORDS OF 4 BYTES
(0.15 % OF TOTAL MEMORY)

R1 = 0.3442E-01
R2 = 0.1226
R3 = 0.1226
X = 0.6569
R = 1.134
N1 = 0.1809
N2 = 0.2819
N3 = 0.2819

1

NUMERO DE USUARIOS= 7

- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*   NAME   * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
```



```

*           *           *           *           *           *
*CPU        *0.3000E-01*0.1830    *0.2161    *0.3543E-01* 6.100    *
*           *           *           *           *           *
*DISC  1    *0.1000    *0.2669    *0.3421    *0.1282    * 2.669    *
*           *           *           *           *           *
*DISC  2    *0.1000    *0.2669    *0.3421    *0.1282    * 2.669    *
*           *           *           *           *           *
*TERMINAL  * 8.000    *0.0000E+00* 6.100    * 8.000    *0.7625    *
*           *           *           *           *           *

```

```

*****

```

MEMORY USED: 7532 WORDS OF 4 BYTES

(0.15 % OF TOTAL MEMORY)

```

R1 = 0.3543E-01
R2 = 0.1282
R3 = 0.1282
X = 0.7625
R = 1.181
N1 = 0.2161
N2 = 0.3421
N3 = 0.3421

```

1

NUMERO DE USUARIOS= 8
- MEAN VALUE ANALYSIS ("MVA") -

```

*****

```

```

* NAME * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*           *           *           *           *           *
*CPU        *0.3000E-01*0.2080    *0.2529    *0.3648E-01* 6.933    *
*           *           *           *           *           *
*DISC  1    *0.1000    *0.3033    *0.4071    *0.1342    * 3.033    *
*           *           *           *           *           *
*DISC  2    *0.1000    *0.3033    *0.4071    *0.1342    * 3.033    *
*           *           *           *           *           *
*TERMINAL  * 8.000    *0.0000E+00* 6.933    * 8.000    *0.8666    *
*           *           *           *           *           *

```

```

*****

```

MEMORY USED: 7555 WORDS OF 4 BYTES

(0.15 % OF TOTAL MEMORY)

```

R1 = 0.3648E-01
R2 = 0.1342
R3 = 0.1342
X = 0.8666
R = 1.231
N1 = 0.2529
N2 = 0.4071
N3 = 0.4071

```

1

NUMERO DE USUARIOS= 9
- MEAN VALUE ANALYSIS ("MVA") -

```

*****

```

```

* NAME * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*           *           *           *           *           *
*CPU        *0.3000E-01*0.2326    *0.2915    *0.3759E-01* 7.754    *
*           *           *           *           *           *
*DISC  1    *0.1000    *0.3392    *0.4773    *0.1407    * 3.392    *

```

```

*          *          *          *          *          *          *
*DISC    2  *0.1000    *0.3392    *0.4773    *0.1407    * 3.392    *
*          *          *          *          *          *          *
*TERMINAL * 8.000     *0.0000E+00* 7.754    * 8.000     *0.9692    *
*          *          *          *          *          *          *
*****

```

```

MEMORY USED:      7578 WORDS OF 4 BYTES
( 0.15 % OF TOTAL MEMORY)

```

```

R1 = 0.3759E-01
R2 = 0.1407
R3 = 0.1407
X = 0.9692
R = 1.286
N1 = 0.2915
N2 = 0.4773
N3 = 0.4773

```

1

```

NUMERO DE USUARIOS=      10
- MEAN VALUE ANALYSIS ("MVA") -

```

```

*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *          *
*CPU       *0.3000E-01*0.2568    *0.3317    *0.3874E-01* 8.562    *
*          *          *          *          *          *          *
*DISC    1  *0.1000    *0.3746    *0.5534    *0.1477    * 3.746    *
*          *          *          *          *          *          *
*DISC    2  *0.1000    *0.3746    *0.5534    *0.1477    * 3.746    *
*          *          *          *          *          *          *
*TERMINAL * 8.000     *0.0000E+00* 8.562    * 8.000     * 1.070    *
*          *          *          *          *          *          *
*****

```

```

MEMORY USED:      7601 WORDS OF 4 BYTES
( 0.15 % OF TOTAL MEMORY)

```

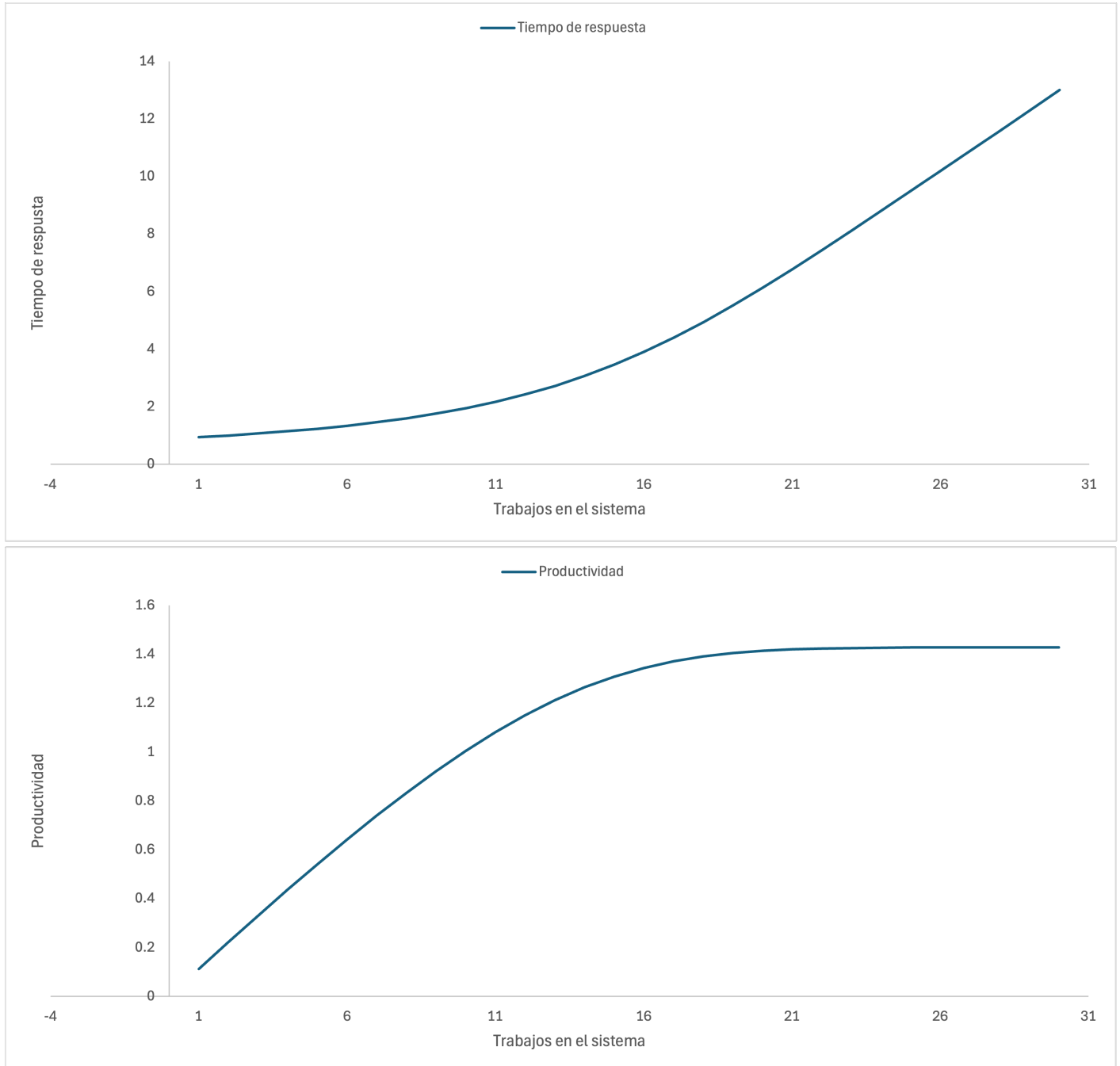
```

R1 = 0.3874E-01
R2 = 0.1477
R3 = 0.1477
X = 1.070
R = 1.344
N1 = 0.3317
N2 = 0.5534
N3 = 0.5534

```

49 /END/

1.5 Modelo d



```

1 /DECLARE/ QUEUE CPU,DISC,TERMINAL;
2     REAL Z=8.;
3     REAL V_CPU=8.,V_DISC=7.;
4     REAL X,X1,X2,R,R1,R2,N1,N2;
5     INTEGER N;
6 /STATION/ NAME=CPU;
7 &     SCHED=PS;
8     SERVICE=EXP(0.03);
9     TRANSIT=DISC,V_DISC,TERMINAL,1;
10 /STATION/ NAME=DISC;
11     TRANSIT=CPU;
12     SERVICE=EXP(0.1);
13 /STATION/ NAME=TERMINAL;

```

```

14      TYPE=INFINITE;
15      INIT=N;
16      SERVICE=EXP(Z);
17      TRANSIT=CPU;
18 /CONTROL/ CLASS=ALL QUEUE;
19 /EXEC/    FOR N:=1 STEP 1 UNTIL 30 DO
20      BEGIN
21          PRINT;
22          PRINT("NUMERO DE USUARIOS=",N);
23          SOLVE;
24
25          X := MTHRUPUT(TERMINAL);
26          X1 := MTHRUPUT(CPU);
27          X2 := MTHRUPUT(DISC);
28          R := N / X - Z;
29          R1 := MRESPONS(CPU);
30          R2 := MRESPONS(DISC);
31          N1 := X1 * R1;
32          N2 := X2 * R2;
33
34          PRINT("R1 = ", R1);
35          PRINT("R2 = ", R2);
36          PRINT("X = ", X);
37          PRINT("R = ", R);
38          PRINT("N1 = ", N1);
39          PRINT("N2 = ", N2);
40      END;

```

```

1
NUMERO DE USUARIOS=      1
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.2685E-01*0.2685E-01*0.3000E-01*0.8949 *
*          *          *          *          *          *
*DISC      *0.1000   *0.7830E-01*0.7830E-01*0.1000   *0.7830 *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00*0.8949   * 8.000   *0.1119 *
*          *          *          *          *          *
*****
          MEMORY USED:      7012 WORDS OF 4 BYTES
          ( 0.14 % OF TOTAL MEMORY)
R1 = 0.3000E-01
R2 = 0.1000
X = 0.1119
R = 0.9400
N1 = 0.2685E-01
N2 = 0.7830E-01

```

```

1
NUMERO DE USUARIOS=      2
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *

```

```

*CPU      *0.3000E-01*0.5333E-01*0.5476E-01*0.3081E-01* 1.778  *
*          *          *          *          *          *          *
*DISC     *0.1000    *0.1555    *0.1677    *0.1078    * 1.555  *
*          *          *          *          *          *          *
*TERMINAL * 8.000    *0.0000E+00* 1.778    * 8.000    *0.2222  *
*          *          *          *          *          *          *

```

```

*****

```

MEMORY USED: 7038 WORDS OF 4 BYTES

(0.14 % OF TOTAL MEMORY)

R1 = 0.3081E-01

R2 = 0.1078

X = 0.2222

R = 1.001

N1 = 0.5476E-01

N2 = 0.1677

1

NUMERO DE USUARIOS= 3

- MEAN VALUE ANALYSIS ("MVA") -

```

*****

```

```

*  NAME  *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *          *
*CPU      *0.3000E-01*0.7938E-01*0.8372E-01*0.3164E-01* 2.646  *
*          *          *          *          *          *          *
*DISC     *0.1000    *0.2315    *0.2703    *0.1168    * 2.315  *
*          *          *          *          *          *          *
*TERMINAL * 8.000    *0.0000E+00* 2.646    * 8.000    *0.3307  *
*          *          *          *          *          *          *

```

```

*****

```

MEMORY USED: 7064 WORDS OF 4 BYTES

(0.14 % OF TOTAL MEMORY)

R1 = 0.3164E-01

R2 = 0.1168

X = 0.3307

R = 1.071

N1 = 0.8372E-01

N2 = 0.2703

1

NUMERO DE USUARIOS= 4

- MEAN VALUE ANALYSIS ("MVA") -

```

*****

```

```

*  NAME  *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *          *
*CPU      *0.3000E-01*0.1049    *0.1137    *0.3251E-01* 3.498  *
*          *          *          *          *          *          *
*DISC     *0.1000    *0.3060    *0.3888    *0.1270    * 3.060  *
*          *          *          *          *          *          *
*TERMINAL * 8.000    *0.0000E+00* 3.498    * 8.000    *0.4372  *
*          *          *          *          *          *          *

```

```

*****

```

MEMORY USED: 7090 WORDS OF 4 BYTES

(0.14 % OF TOTAL MEMORY)

R1 = 0.3251E-01

R2 = 0.1270

X = 0.4372

R = 1.149
 N1 = 0.1137
 N2 = 0.3888

1

```

NUMERO DE USUARIOS=      5
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *
*CPU       *0.3000E-01*0.1299  *0.1446  *0.3341E-01* 4.329  *
*          *         *         *         *         *
*DISC      *0.1000   *0.3788   *0.5261  *0.1389   * 3.788  *
*          *         *         *         *         *
*TERMINAL  * 8.000   *0.0000E+00* 4.329  * 8.000   *0.5412  *
*          *         *         *         *         *
*****
      MEMORY USED:      7116 WORDS OF 4 BYTES
      ( 0.14 % OF TOTAL MEMORY)
R1 = 0.3341E-01
R2 = 0.1389
X = 0.5412
R = 1.239
N1 = 0.1446
N2 = 0.5261

```

1

```

NUMERO DE USUARIOS=      6
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *
*CPU       *0.3000E-01*0.1541  *0.1764  *0.3434E-01* 5.138  *
*          *         *         *         *         *
*DISC      *0.1000   *0.4495   *0.6860  *0.1526   * 4.495  *
*          *         *         *         *         *
*TERMINAL  * 8.000   *0.0000E+00* 5.138  * 8.000   *0.6422  *
*          *         *         *         *         *
*****
      MEMORY USED:      7142 WORDS OF 4 BYTES
      ( 0.14 % OF TOTAL MEMORY)
R1 = 0.3434E-01
R2 = 0.1526
X = 0.6422
R = 1.343
N1 = 0.1764
N2 = 0.6860

```

1

```

NUMERO DE USUARIOS=      7
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *
*CPU       *0.3000E-01*0.1775  *0.2089  *0.3529E-01* 5.918  *

```

```

*          *          *          *          *          *          *
*DISC      *0.1000    *0.5178    *0.8731    *0.1686    * 5.178    *
*          *          *          *          *          *          *
*TERMINAL  * 8.000    *0.0000E+00* 5.918    * 8.000    *0.7398    *
*          *          *          *          *          *          *
*****
      MEMORY USED:      7168 WORDS OF 4 BYTES
      ( 0.14 % OF TOTAL MEMORY)

R1 = 0.3529E-01
R2 = 0.1686
X = 0.7398
R = 1.463
N1 = 0.2089
N2 = 0.8731

```

1

```

NUMERO DE USUARIOS=      8
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *          *
*CPU       *0.3000E-01*0.2000    *0.2417    *0.3627E-01* 6.666    *
*          *          *          *          *          *          *
*DISC      *0.1000    *0.5833    * 1.092    *0.1873    * 5.833    *
*          *          *          *          *          *          *
*TERMINAL  * 8.000    *0.0000E+00* 6.666    * 8.000    *0.8332    *
*          *          *          *          *          *          *
*****
      MEMORY USED:      7194 WORDS OF 4 BYTES
      ( 0.14 % OF TOTAL MEMORY)

R1 = 0.3627E-01
R2 = 0.1873
X = 0.8332
R = 1.601
N1 = 0.2417
N2 = 1.092

```

1

```

NUMERO DE USUARIOS=      9
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *          *
*CPU       *0.3000E-01*0.2212    *0.2747    *0.3725E-01* 7.375    *
*          *          *          *          *          *          *
*DISC      *0.1000    *0.6453    * 1.350    *0.2092    * 6.453    *
*          *          *          *          *          *          *
*TERMINAL  * 8.000    *0.0000E+00* 7.375    * 8.000    *0.9219    *
*          *          *          *          *          *          *
*****
      MEMORY USED:      7220 WORDS OF 4 BYTES
      ( 0.14 % OF TOTAL MEMORY)

R1 = 0.3725E-01
R2 = 0.2092
X = 0.9219
R = 1.763

```

N1 = 0.2747
N2 = 1.350

1

```

NUMERO DE USUARIOS=      10
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *          *
*CPU       *0.3000E-01*0.2412   *0.3074   *0.3824E-01* 8.039   *
*          *          *          *          *          *          *
*DISC      *0.1000   *0.7034   * 1.653   *0.2350   * 7.034   *
*          *          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 8.039   * 8.000   * 1.005   *
*          *          *          *          *          *          *
*****
      MEMORY USED:      7246 WORDS OF 4 BYTES
      ( 0.14 % OF TOTAL MEMORY)

R1 = 0.3824E-01
R2 = 0.2350
X = 1.005
R = 1.951
N1 = 0.3074
N2 = 1.653

```

1

```

NUMERO DE USUARIOS=      11
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *          *
*CPU       *0.3000E-01*0.2596   *0.3394   *0.3922E-01* 8.652   *
*          *          *          *          *          *          *
*DISC      *0.1000   *0.7570   * 2.009   *0.2653   * 7.570   *
*          *          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 8.652   * 8.000   * 1.081   *
*          *          *          *          *          *          *
*****
      MEMORY USED:      7272 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

R1 = 0.3922E-01
R2 = 0.2653
X = 1.081
R = 2.171
N1 = 0.3394
N2 = 2.009

```

1

```

NUMERO DE USUARIOS=      12
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *          *
*CPU       *0.3000E-01*0.2762   *0.3699   *0.4018E-01* 9.206   *
*          *          *          *          *          *          *

```



```
*DISC      *0.1000    *0.8056    * 2.424    *0.3009    * 8.056    *
*          *          *          *          *          *          *
*TERMINAL  * 8.000    *0.0000E+00* 9.206    * 8.000    * 1.151    *
*          *          *          *          *          *          *
```

```
*****
MEMORY USED:      7298 WORDS OF 4 BYTES
( 0.15 % OF TOTAL MEMORY)
```

```
R1 = 0.4018E-01
R2 = 0.3009
X = 1.151
R = 2.428
N1 = 0.3699
N2 = 2.424
```

1

```
NUMERO DE USUARIOS=      13
- MEAN VALUE ANALYSIS ("MVA") -
```

```
*****
* NAME      * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *
*CPU        *0.3000E-01*0.2909    *0.3985    *0.4110E-01* 9.697    *
*          *          *          *          *          *
*DISC       *0.1000    *0.8485    * 2.905    *0.3424    * 8.485    *
*          *          *          *          *          *
*TERMINAL   * 8.000    *0.0000E+00* 9.697    * 8.000    * 1.212    *
*          *          *          *          *          *
```

```
*****
MEMORY USED:      7324 WORDS OF 4 BYTES
( 0.15 % OF TOTAL MEMORY)
```

```
R1 = 0.4110E-01
R2 = 0.3424
X = 1.212
R = 2.725
N1 = 0.3985
N2 = 2.905
```

1

```
NUMERO DE USUARIOS=      14
- MEAN VALUE ANALYSIS ("MVA") -
```

```
*****
* NAME      * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *
*CPU        *0.3000E-01*0.3035    *0.4245    *0.4196E-01* 10.12    *
*          *          *          *          *          *
*DISC       *0.1000    *0.8854    * 3.457    *0.3905    * 8.854    *
*          *          *          *          *          *
*TERMINAL   * 8.000    *0.0000E+00* 10.12    * 8.000    * 1.265    *
*          *          *          *          *          *
```

```
*****
MEMORY USED:      7350 WORDS OF 4 BYTES
( 0.15 % OF TOTAL MEMORY)
```

```
R1 = 0.4196E-01
R2 = 0.3905
X = 1.265
R = 3.069
N1 = 0.4245
```

N2 = 3.457

1

```

NUMERO DE USUARIOS=      15
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.3141   *0.4474   *0.4274E-01* 10.47   *
*          *          *          *          *          *
*DISC      *0.1000   *0.9161   * 4.083   *0.4457   * 9.161   *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 10.47   * 8.000   * 1.309   *
*          *          *          *          *          *
*****
      MEMORY USED:      7376 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

R1 = 0.4274E-01
R2 = 0.4457
X = 1.309
R = 3.462
N1 = 0.4474
N2 = 4.083

```

1

```

NUMERO DE USUARIOS=      16
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.3225   *0.4668   *0.4342E-01* 10.75   *
*          *          *          *          *          *
*DISC      *0.1000   *0.9407   * 4.782   *0.5083   * 9.407   *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 10.75   * 8.000   * 1.344   *
*          *          *          *          *          *
*****
      MEMORY USED:      7402 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

R1 = 0.4342E-01
R2 = 0.5083
X = 1.344
R = 3.906
N1 = 0.4668
N2 = 4.782

```

1

```

NUMERO DE USUARIOS=      17
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.3290   *0.4827   *0.4401E-01* 10.97   *
*          *          *          *          *          *
*DISC      *0.1000   *0.9597   * 5.549   *0.5782   * 9.597   *

```

```

*           *           *           *           *           *           *
*TERMINAL * 8.000      *0.0000E+00* 10.97    * 8.000      * 1.371    *
*           *           *           *           *           *           *
*****
      MEMORY USED:      7428 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

R1 = 0.4401E-01
R2 = 0.5782
X = 1.371
R = 4.399
N1 = 0.4827
N2 = 5.549

```

1

```

NUMERO DE USUARIOS=      18
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*           *           *           *           *           *           *
*CPU       *0.3000E-01*0.3338   *0.4950   *0.4448E-01* 11.13   *
*           *           *           *           *           *           *
*DISC      *0.1000   *0.9737   * 6.377   *0.6549   * 9.737   *
*           *           *           *           *           *           *
*TERMINAL  * 8.000   *0.0000E+00* 11.13   * 8.000   * 1.391   *
*           *           *           *           *           *           *
*****
      MEMORY USED:      7454 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

R1 = 0.4448E-01
R2 = 0.6549
X = 1.391
R = 4.940
N1 = 0.4950
N2 = 6.377

```

1

```

NUMERO DE USUARIOS=      19
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*           *           *           *           *           *           *
*CPU       *0.3000E-01*0.3372   *0.5041   *0.4485E-01* 11.24   *
*           *           *           *           *           *           *
*DISC      *0.1000   *0.9835   * 7.255   *0.7377   * 9.835   *
*           *           *           *           *           *           *
*TERMINAL  * 8.000   *0.0000E+00* 11.24   * 8.000   * 1.405   *
*           *           *           *           *           *           *
*****
      MEMORY USED:      7480 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

R1 = 0.4485E-01
R2 = 0.7377
X = 1.405
R = 5.523
N1 = 0.5041
N2 = 7.255

```

1

```

NUMERO DE USUARIOS=      20
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.3395   *0.5106   *0.4512E-01* 11.32  *
*          *          *          *          *          *
*DISC      *0.1000   *0.9901   * 8.174   *0.8255   * 9.901  *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 11.32   * 8.000   * 1.414  *
*          *          *          *          *          *
*****
          MEMORY USED:      7506 WORDS OF 4 BYTES
          ( 0.15 % OF TOTAL MEMORY)

R1 = 0.4512E-01
R2 = 0.8255
X = 1.414
R = 6.140
N1 = 0.5106
N2 = 8.174

```

1

```

NUMERO DE USUARIOS=      21
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.3409   *0.5150   *0.4532E-01* 11.36  *
*          *          *          *          *          *
*DISC      *0.1000   *0.9943   * 9.122   *0.9174   * 9.943  *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 11.36   * 8.000   * 1.420  *
*          *          *          *          *          *
*****
          MEMORY USED:      7532 WORDS OF 4 BYTES
          ( 0.15 % OF TOTAL MEMORY)

R1 = 0.4532E-01
R2 = 0.9174
X = 1.420
R = 6.784
N1 = 0.5150
N2 = 9.122

```

1

```

NUMERO DE USUARIOS=      22
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.3418   *0.5178   *0.4545E-01* 11.39  *
*          *          *          *          *          *
*DISC      *0.1000   *0.9968   * 10.09   * 1.012   * 9.968  *
*          *          *          *          *          *

```

```

*TERMINAL * 8.000 *0.0000E+00* 11.39 * 8.000 * 1.424 *
* * * * *
*****
MEMORY USED: 7558 WORDS OF 4 BYTES
( 0.15 % OF TOTAL MEMORY)
R1 = 0.4545E-01
R2 = 1.012
X = 1.424
R = 7.449
N1 = 0.5178
N2 = 10.09

```

1

```

NUMERO DE USUARIOS= 23
- MEAN VALUE ANALYSIS ("MVA") -
*****
* NAME * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
* * * * *
*CPU *0.3000E-01*0.3423 *0.5195 *0.4553E-01* 11.41 *
* * * * *
*DISC *0.1000 *0.9983 * 11.07 * 1.109 * 9.983 *
* * * * *
*TERMINAL * 8.000 *0.0000E+00* 11.41 * 8.000 * 1.426 *
* * * * *
*****
MEMORY USED: 7584 WORDS OF 4 BYTES
( 0.15 % OF TOTAL MEMORY)
R1 = 0.4553E-01
R2 = 1.109
X = 1.426
R = 8.127
N1 = 0.5195
N2 = 11.07

```

1

```

NUMERO DE USUARIOS= 24
- MEAN VALUE ANALYSIS ("MVA") -
*****
* NAME * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
* * * * *
*CPU *0.3000E-01*0.3426 *0.5205 *0.4559E-01* 11.42 *
* * * * *
*DISC *0.1000 *0.9991 * 12.06 * 1.207 * 9.991 *
* * * * *
*TERMINAL * 8.000 *0.0000E+00* 11.42 * 8.000 * 1.427 *
* * * * *
*****
MEMORY USED: 7610 WORDS OF 4 BYTES
( 0.15 % OF TOTAL MEMORY)
R1 = 0.4559E-01
R2 = 1.207
X = 1.427
R = 8.814
N1 = 0.5205
N2 = 12.06

```

```

1
NUMERO DE USUARIOS=      25
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *          *
*CPU       *0.3000E-01*0.3427   *0.5211   *0.4562E-01* 11.42   *
*          *          *          *          *          *          *
*DISC      *0.1000   *0.9996   * 13.06   * 1.306   * 9.996   *
*          *          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 11.42   * 8.000   * 1.428   *
*          *          *          *          *          *          *
*****
          MEMORY USED:      7636 WORDS OF 4 BYTES
          ( 0.15 % OF TOTAL MEMORY)

R1 = 0.4562E-01
R2 = 1.306
X = 1.428
R = 9.507
N1 = 0.5211
N2 = 13.06

```

```

1
NUMERO DE USUARIOS=      26
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *          *
*CPU       *0.3000E-01*0.3428   *0.5214   *0.4563E-01* 11.43   *
*          *          *          *          *          *          *
*DISC      *0.1000   *0.9998   * 14.05   * 1.406   * 9.998   *
*          *          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 11.43   * 8.000   * 1.428   *
*          *          *          *          *          *          *
*****
          MEMORY USED:      7662 WORDS OF 4 BYTES
          ( 0.15 % OF TOTAL MEMORY)

R1 = 0.4563E-01
R2 = 1.406
X = 1.428
R = 10.20
N1 = 0.5214
N2 = 14.05

```

```

1
NUMERO DE USUARIOS=      27
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *          *
*CPU       *0.3000E-01*0.3428   *0.5216   *0.4564E-01* 11.43   *
*          *          *          *          *          *          *
*DISC      *0.1000   *0.9999   * 15.05   * 1.505   * 9.999   *
*          *          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 11.43   * 8.000   * 1.428   *
*          *          *          *          *          *          *
*****

```

```

*           *           *           *           *           *           *
*****
      MEMORY USED:      7688 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

R1 = 0.4564E-01
R2 = 1.505
X = 1.428
R = 10.90
N1 = 0.5216
N2 = 15.05

```

1

```

NUMERO DE USUARIOS=      28
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME  *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*           *           *           *           *           *           *
*CPU      *0.3000E-01*0.3428   *0.5217   *0.4565E-01* 11.43   *
*           *           *           *           *           *           *
*DISC     *0.1000   * 1.000   * 16.05   * 1.605   *10.000   *
*           *           *           *           *           *           *
*TERMINAL * 8.000   *0.0000E+00* 11.43   * 8.000   * 1.429   *
*           *           *           *           *           *           *
*****
      MEMORY USED:      7714 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

```

```

R1 = 0.4565E-01
R2 = 1.605
X = 1.429
R = 11.60
N1 = 0.5217
N2 = 16.05

```

1

```

NUMERO DE USUARIOS=      29
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME  *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*           *           *           *           *           *           *
*CPU      *0.3000E-01*0.3429   *0.5217   *0.4565E-01* 11.43   *
*           *           *           *           *           *           *
*DISC     *0.1000   * 1.000   * 17.05   * 1.705   *10.000   *
*           *           *           *           *           *           *
*TERMINAL * 8.000   *0.0000E+00* 11.43   * 8.000   * 1.429   *
*           *           *           *           *           *           *
*****
      MEMORY USED:      7740 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

```

```

R1 = 0.4565E-01
R2 = 1.705
X = 1.429
R = 12.30
N1 = 0.5217
N2 = 17.05

```

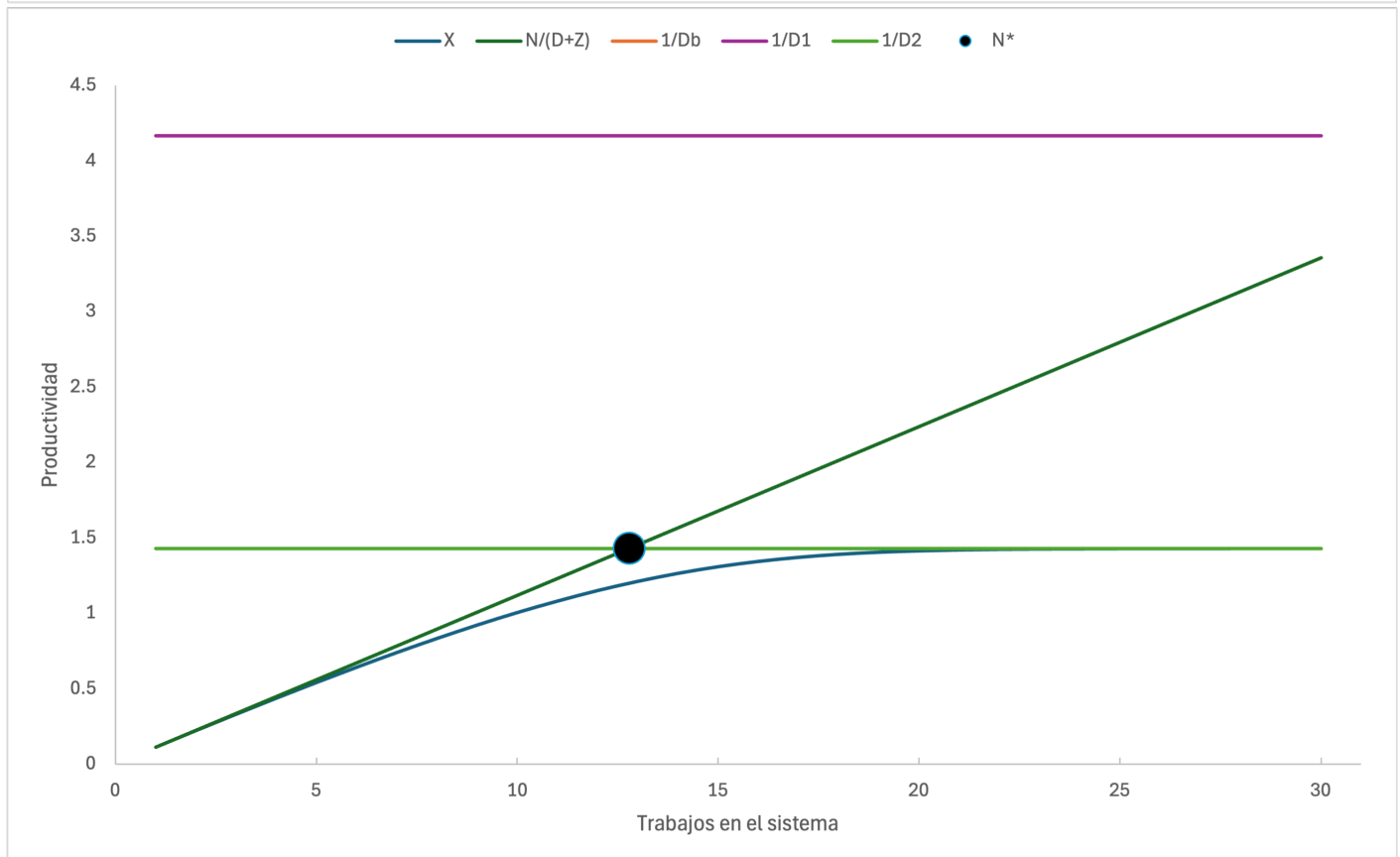
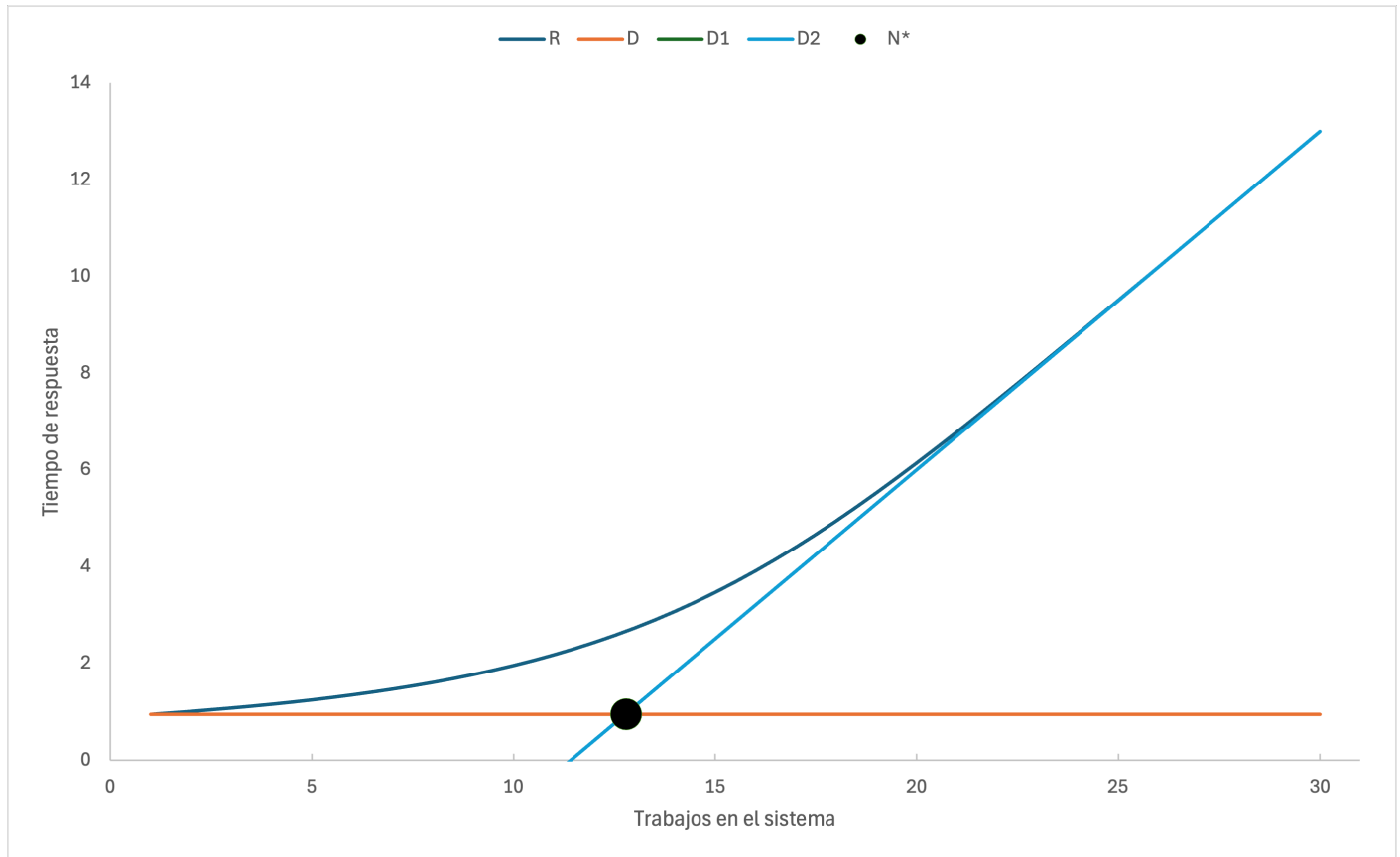
1

```

NUMERO DE USUARIOS=      30
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.3429   *0.5217   *0.4565E-01* 11.43  *
*          *          *          *          *          *
*DISC      *0.1000   * 1.000   * 18.05   * 1.805   *10.000  *
*          *          *          *          *          *
*TERMINAL  * 8.000   *0.0000E+00* 11.43   * 8.000   * 1.429  *
*          *          *          *          *          *
*****
          MEMORY USED:      7766 WORDS OF 4 BYTES
          ( 0.16  % OF TOTAL MEMORY)
R1 = 0.4565E-01
R2 = 1.805
X = 1.429
R = 13.00
N1 = 0.5217
N2 = 18.05
41 /END/

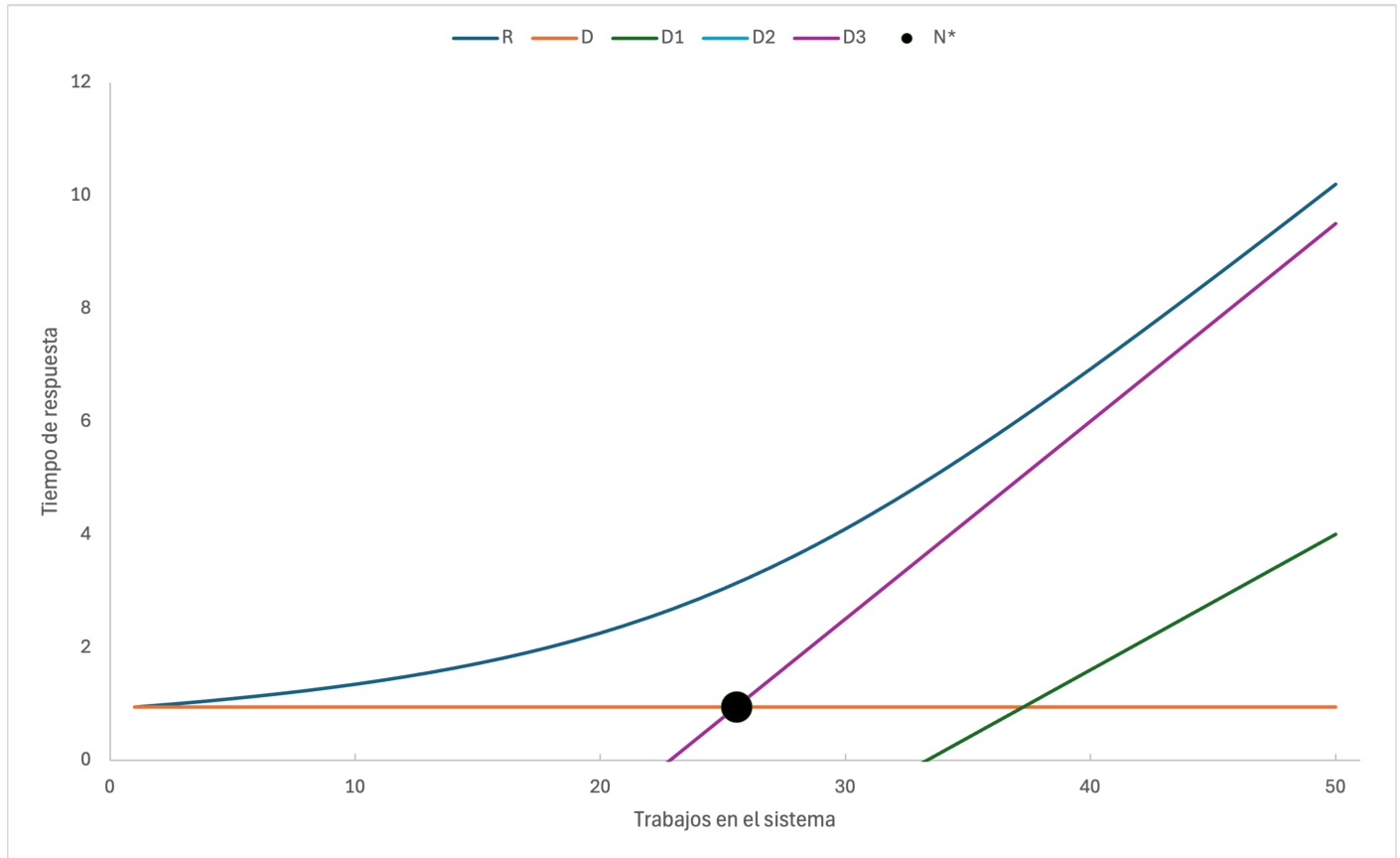
```

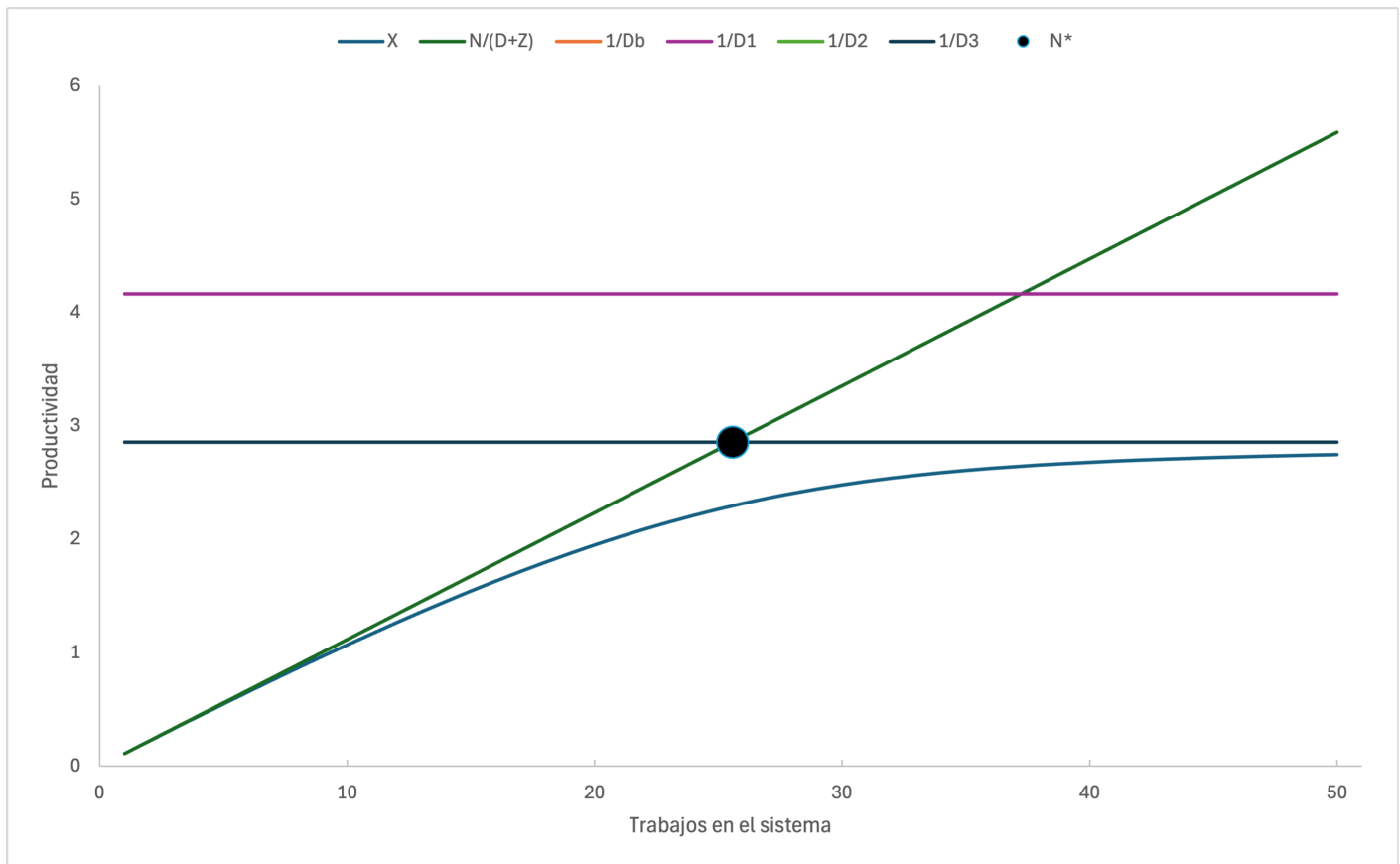

1.6 Modelo e



1.7 Modelo f

He incrementado el numero de iteraciones a 50 para poder visualizar mejor la convergencia del tiempo de respuesta con los límites asintóticos.





```

1 /DECLARE/ QUEUE CPU,DISC(2),TERMINAL;
2     REAL Z=8.;
3     REAL V_CPU=8.,V_DISC(2)=(3.5,3.5);
4     REAL X,X1,X2,X3,R,R1,R2,R3,N1,N2,N3;
5     INTEGER N;
6 /STATION/ NAME=CPU;
7 &     SCHED=PS;
8     SERVICE=EXP(0.03);
9     TRANSIT=DISC,V_DISC,TERMINAL,1;
10 /STATION/ NAME=DISC;
11     TRANSIT=CPU;
12 /STATION/ NAME=DISC(1);
13     SERVICE=EXP(0.1);
14 /STATION/ NAME=DISC(2);
15     SERVICE=EXP(0.1);
16 /STATION/ NAME=TERMINAL;
17     TYPE=INFINITE;
18     INIT=N;
19     SERVICE=EXP(Z);
20     TRANSIT=CPU;
21 /CONTROL/ CLASS=ALL QUEUE;
22 /EXEC/   FOR N:=1 STEP 1 UNTIL 50 DO
23     BEGIN
24         PRINT;
25         PRINT("NUMERO DE USUARIOS=",N);
26         SOLVE;
27
28         X := MTHRUPUT(TERMINAL);
29         X1 := MTHRUPUT(CPU);
30         X2 := MTHRUPUT(DISC(1));

```

```

31      X3 := MTHRUPUT(DISC(2));
32      R := N / X - Z;
33      R1 := MRESPONS(CPU);
34      R2 := MRESPONS(DISC(1));
35      R3 := MRESPONS(DISC(2));
36      N1 := X1 * R1;
37      N2 := X2 * R2;
38      N3 := X2 * R3;
39
40      PRINT("R1 = ", R1);
41      PRINT("R2 = ", R2);
42      PRINT("R3 = ", R3);
43      PRINT("X = ", X);
44      PRINT("R = ", R);
45      PRINT("N1 = ", N1);
46      PRINT("N2 = ", N2);
47      PRINT("N3 = ", N3);
48      END;

```

1

```

NUMERO DE USUARIOS=      1
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.2685E-01*0.2685E-01*0.3000E-01*0.8949 *
*          *          *          *          *          *
*DISC  1   *0.1000   *0.3915E-01*0.3915E-01*0.1000   *0.3915 *
*          *          *          *          *          *
*DISC  2   *0.1000   *0.3915E-01*0.3915E-01*0.1000   *0.3915 *
*          *          *          *          *          *
*TERMINAL * 8.000    *0.0000E+00*0.8949   * 8.000    *0.1119 *
*          *          *          *          *          *
*****
      MEMORY USED:      7434 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)
R1 =  0.3000E-01
R2 =  0.1000
R3 =  0.1000
X =  0.1119
R =  0.9400
N1 =  0.2685E-01
N2 =  0.3915E-01
N3 =  0.3915E-01

```

1

```

NUMERO DE USUARIOS=      2
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.5349E-01*0.5492E-01*0.3081E-01* 1.783 *
*          *          *          *          *          *
*DISC  1   *0.1000   *0.7800E-01*0.8106E-01*0.1039   *0.7800 *
*          *          *          *          *          *
*DISC  2   *0.1000   *0.7800E-01*0.8106E-01*0.1039   *0.7800 *

```

```

*           *           *           *           *           *           *
*TERMINAL * 8.000      *0.0000E+00* 1.783      * 8.000      *0.2229      *
*           *           *           *           *           *           *
*****
      MEMORY USED:      7457 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

R1 = 0.3081E-01
R2 = 0.1039
R3 = 0.1039
X = 0.2229
R = 0.9738
N1 = 0.5492E-01
N2 = 0.8106E-01
N3 = 0.8106E-01

```

1

```

NUMERO DE USUARIOS=      3
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*           *           *           *           *           *           *
*CPU       *0.3000E-01*0.7991E-01*0.8430E-01*0.3165E-01* 2.664      *
*           *           *           *           *           *           *
*DISC  1   *0.1000      *0.1165      *0.1260      *0.1081      * 1.165      *
*           *           *           *           *           *           *
*DISC  2   *0.1000      *0.1165      *0.1260      *0.1081      * 1.165      *
*           *           *           *           *           *           *
*TERMINAL * 8.000      *0.0000E+00* 2.664      * 8.000      *0.3330      *
*           *           *           *           *           *           *
*****
      MEMORY USED:      7480 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

R1 = 0.3165E-01
R2 = 0.1081
R3 = 0.1081
X = 0.3330
R = 1.010
N1 = 0.8430E-01
N2 = 0.1260
N3 = 0.1260

```

1

```

NUMERO DE USUARIOS=      4
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*           *           *           *           *           *           *
*CPU       *0.3000E-01*0.1061      *0.1150      *0.3253E-01* 3.537      *
*           *           *           *           *           *           *
*DISC  1   *0.1000      *0.1547      *0.1742      *0.1126      * 1.547      *
*           *           *           *           *           *           *
*DISC  2   *0.1000      *0.1547      *0.1742      *0.1126      * 1.547      *
*           *           *           *           *           *           *
*TERMINAL * 8.000      *0.0000E+00* 3.537      * 8.000      *0.4421      *
*           *           *           *           *           *           *
*****

```

MEMORY USED: 7503 WORDS OF 4 BYTES
(0.15 % OF TOTAL MEMORY)

R1 = 0.3253E-01
R2 = 0.1126
R3 = 0.1126
X = 0.4421
R = 1.048
N1 = 0.1150
N2 = 0.1742
N3 = 0.1742

1

NUMERO DE USUARIOS= 5
- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*  NAME  * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *
*CPU      *0.3000E-01*0.1320  *0.1472  *0.3345E-01* 4.401  *
*          *          *          *          *          *
*DISC  1  *0.1000  *0.1925  *0.2261  *0.1174  * 1.925  *
*          *          *          *          *          *
*DISC  2  *0.1000  *0.1925  *0.2261  *0.1174  * 1.925  *
*          *          *          *          *          *
*TERMINAL * 8.000   *0.0000E+00* 4.401  * 8.000   *0.5501  *
*          *          *          *          *          *
```

MEMORY USED: 7526 WORDS OF 4 BYTES
(0.15 % OF TOTAL MEMORY)

R1 = 0.3345E-01
R2 = 0.1174
R3 = 0.1174
X = 0.5501
R = 1.090
N1 = 0.1472
N2 = 0.2261
N3 = 0.2261

1

NUMERO DE USUARIOS= 6
- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*  NAME  * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *
*CPU      *0.3000E-01*0.1577  *0.1809  *0.3442E-01* 5.255  *
*          *          *          *          *          *
*DISC  1  *0.1000  *0.2299  *0.2819  *0.1226  * 2.299  *
*          *          *          *          *          *
*DISC  2  *0.1000  *0.2299  *0.2819  *0.1226  * 2.299  *
*          *          *          *          *          *
*TERMINAL * 8.000   *0.0000E+00* 5.255  * 8.000   *0.6569  *
*          *          *          *          *          *
```

MEMORY USED: 7549 WORDS OF 4 BYTES
(0.15 % OF TOTAL MEMORY)

R1 = 0.3442E-01
R2 = 0.1226

R3 = 0.1226
 X = 0.6569
 R = 1.134
 N1 = 0.1809
 N2 = 0.2819
 N3 = 0.2819

1

NUMERO DE USUARIOS= 7
 - MEAN VALUE ANALYSIS ("MVA") -

 * NAME * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *

 * * * * *
 *CPU *0.3000E-01*0.1830 *0.2161 *0.3543E-01* 6.100 *
 * * * * *
 *DISC 1 *0.1000 *0.2669 *0.3421 *0.1282 * 2.669 *
 * * * * *
 *DISC 2 *0.1000 *0.2669 *0.3421 *0.1282 * 2.669 *
 * * * * *
 *TERMINAL * 8.000 *0.0000E+00* 6.100 * 8.000 *0.7625 *
 * * * * *

 MEMORY USED: 7572 WORDS OF 4 BYTES
 (0.15 % OF TOTAL MEMORY)

R1 = 0.3543E-01
 R2 = 0.1282
 R3 = 0.1282
 X = 0.7625
 R = 1.181
 N1 = 0.2161
 N2 = 0.3421
 N3 = 0.3421

1

NUMERO DE USUARIOS= 8
 - MEAN VALUE ANALYSIS ("MVA") -

 * NAME * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *

 * * * * *
 *CPU *0.3000E-01*0.2080 *0.2529 *0.3648E-01* 6.933 *
 * * * * *
 *DISC 1 *0.1000 *0.3033 *0.4071 *0.1342 * 3.033 *
 * * * * *
 *DISC 2 *0.1000 *0.3033 *0.4071 *0.1342 * 3.033 *
 * * * * *
 *TERMINAL * 8.000 *0.0000E+00* 6.933 * 8.000 *0.8666 *
 * * * * *

 MEMORY USED: 7595 WORDS OF 4 BYTES
 (0.15 % OF TOTAL MEMORY)

R1 = 0.3648E-01
 R2 = 0.1342
 R3 = 0.1342
 X = 0.8666
 R = 1.231
 N1 = 0.2529

N2 = 0.4071

N3 = 0.4071

1

NUMERO DE USUARIOS= 9

- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *         *
*CPU       *0.3000E-01*0.2326  *0.2915  *0.3759E-01* 7.754  *
*          *         *         *         *         *         *
*DISC  1   *0.1000    *0.3392  *0.4773  *0.1407  * 3.392  *
*          *         *         *         *         *         *
*DISC  2   *0.1000    *0.3392  *0.4773  *0.1407  * 3.392  *
*          *         *         *         *         *         *
*TERMINAL * 8.000     *0.0000E+00* 7.754  * 8.000  *0.9692  *
*          *         *         *         *         *         *
```

MEMORY USED: 7618 WORDS OF 4 BYTES

(0.15 % OF TOTAL MEMORY)

R1 = 0.3759E-01

R2 = 0.1407

R3 = 0.1407

X = 0.9692

R = 1.286

N1 = 0.2915

N2 = 0.4773

N3 = 0.4773

1

NUMERO DE USUARIOS= 10

- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *         *
*CPU       *0.3000E-01*0.2568  *0.3317  *0.3874E-01* 8.562  *
*          *         *         *         *         *         *
*DISC  1   *0.1000    *0.3746  *0.5534  *0.1477  * 3.746  *
*          *         *         *         *         *         *
*DISC  2   *0.1000    *0.3746  *0.5534  *0.1477  * 3.746  *
*          *         *         *         *         *         *
*TERMINAL * 8.000     *0.0000E+00* 8.562  * 8.000  * 1.070  *
*          *         *         *         *         *         *
```

MEMORY USED: 7641 WORDS OF 4 BYTES

(0.15 % OF TOTAL MEMORY)

R1 = 0.3874E-01

R2 = 0.1477

R3 = 0.1477

X = 1.070

R = 1.344

N1 = 0.3317

N2 = 0.5534

N3 = 0.5534

1


```

NUMERO DE USUARIOS=      11
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *          *
*CPU       *0.3000E-01*0.2806   *0.3737   *0.3995E-01*  9.355   *
*          *          *          *          *          *          *
*DISC  1   *0.1000   *0.4093   *0.6357   *0.1553   *  4.093   *
*          *          *          *          *          *          *
*DISC  2   *0.1000   *0.4093   *0.6357   *0.1553   *  4.093   *
*          *          *          *          *          *          *
*TERMINAL * 8.000    *0.0000E+00*  9.355   * 8.000    *  1.169   *
*          *          *          *          *          *          *
*****
      MEMORY USED:      7664 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

R1 =  0.3995E-01
R2 =  0.1553
R3 =  0.1553
X =  1.169
R =  1.407
N1 =  0.3737
N2 =  0.6357
N3 =  0.6357

```

1

```

NUMERO DE USUARIOS=      12
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *          *
*CPU       *0.3000E-01*0.3040   *0.4176   *0.4121E-01* 10.13   *
*          *          *          *          *          *          *
*DISC  1   *0.1000   *0.4433   *0.7251   *0.1636   *  4.433   *
*          *          *          *          *          *          *
*DISC  2   *0.1000   *0.4433   *0.7251   *0.1636   *  4.433   *
*          *          *          *          *          *          *
*TERMINAL * 8.000    *0.0000E+00* 10.13   * 8.000    *  1.267   *
*          *          *          *          *          *          *
*****
      MEMORY USED:      7687 WORDS OF 4 BYTES
      ( 0.15 % OF TOTAL MEMORY)

R1 =  0.4121E-01
R2 =  0.1636
R3 =  0.1636
X =  1.267
R =  1.475
N1 =  0.4176
N2 =  0.7251
N3 =  0.7251

```

1

```

NUMERO DE USUARIOS=      13
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****

```

```
*****
*           *           *           *           *           *
*CPU        *0.3000E-01*0.3268    *0.4632    *0.4253E-01* 10.89 *
*           *           *           *           *           *
*DISC  1    *0.1000    *0.4766    *0.8221    *0.1725    * 4.766 *
*           *           *           *           *           *
*DISC  2    *0.1000    *0.4766    *0.8221    *0.1725    * 4.766 *
*           *           *           *           *           *
*TERMINAL  * 8.000    *0.0000E+00* 10.89    * 8.000    * 1.362 *
*           *           *           *           *           *
*****
```

MEMORY USED: 7710 WORDS OF 4 BYTES
(0.15 % OF TOTAL MEMORY)

R1 = 0.4253E-01
R2 = 0.1725
R3 = 0.1725
X = 1.362
R = 1.548
N1 = 0.4632
N2 = 0.8221
N3 = 0.8221

1

NUMERO DE USUARIOS= 14
- MEAN VALUE ANALYSIS ("MVA") -

```
*****
* NAME      * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*           *           *           *           *           *
*CPU        *0.3000E-01*0.3490    *0.5107    *0.4390E-01* 11.63 *
*           *           *           *           *           *
*DISC  1    *0.1000    *0.5090    *0.9275    *0.1822    * 5.090 *
*           *           *           *           *           *
*DISC  2    *0.1000    *0.5090    *0.9275    *0.1822    * 5.090 *
*           *           *           *           *           *
*TERMINAL  * 8.000    *0.0000E+00* 11.63    * 8.000    * 1.454 *
*           *           *           *           *           *
*****
```

MEMORY USED: 7733 WORDS OF 4 BYTES
(0.15 % OF TOTAL MEMORY)

R1 = 0.4390E-01
R2 = 0.1822
R3 = 0.1822
X = 1.454
R = 1.627
N1 = 0.5107
N2 = 0.9275
N3 = 0.9275

1

NUMERO DE USUARIOS= 15
- MEAN VALUE ANALYSIS ("MVA") -

```
*****
* NAME      * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*           *           *           *           *           *
*CPU        *0.3000E-01*0.3707    *0.5600    *0.4532E-01* 12.36 *
*           *           *           *           *           *
```

```

*DISC  1  *0.1000    *0.5406    * 1.042    *0.1927    * 5.406    *
*          *          *          *          *          *          *
*DISC  2  *0.1000    *0.5406    * 1.042    *0.1927    * 5.406    *
*          *          *          *          *          *          *
*TERMINAL * 8.000    *0.0000E+00* 12.36    * 8.000    * 1.545    *
*          *          *          *          *          *          *
*****

```

```

MEMORY USED:      7756 WORDS OF 4 BYTES
( 0.16 % OF TOTAL MEMORY)

```

```

R1 = 0.4532E-01
R2 = 0.1927
R3 = 0.1927
X = 1.545
R = 1.712
N1 = 0.5600
N2 = 1.042
N3 = 1.042

```

1

```

NUMERO DE USUARIOS=      16
- MEAN VALUE ANALYSIS ("MVA") -

```

```

*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *          *
*CPU       *0.3000E-01*0.3917    *0.6110    *0.4680E-01* 13.06    *
*          *          *          *          *          *          *
*DISC  1  *0.1000    *0.5712    * 1.166    *0.2042    * 5.712    *
*          *          *          *          *          *          *
*DISC  2  *0.1000    *0.5712    * 1.166    *0.2042    * 5.712    *
*          *          *          *          *          *          *
*TERMINAL * 8.000    *0.0000E+00* 13.06    * 8.000    * 1.632    *
*          *          *          *          *          *          *
*****

```

```

MEMORY USED:      7779 WORDS OF 4 BYTES
( 0.16 % OF TOTAL MEMORY)

```

```

R1 = 0.4680E-01
R2 = 0.2042
R3 = 0.2042
X = 1.632
R = 1.804
N1 = 0.6110
N2 = 1.166
N3 = 1.166

```

1

```

NUMERO DE USUARIOS=      17
- MEAN VALUE ANALYSIS ("MVA") -

```

```

*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *          *
*CPU       *0.3000E-01*0.4120    *0.6637    *0.4833E-01* 13.73    *
*          *          *          *          *          *          *
*DISC  1  *0.1000    *0.6008    * 1.302    *0.2166    * 6.008    *
*          *          *          *          *          *          *
*DISC  2  *0.1000    *0.6008    * 1.302    *0.2166    * 6.008    *
*          *          *          *          *          *          *

```

```

*TERMINAL * 8.000 *0.0000E+00* 13.73 * 8.000 * 1.717 *
* * * * *
*****
MEMORY USED: 7802 WORDS OF 4 BYTES
( 0.16 % OF TOTAL MEMORY)
R1 = 0.4833E-01
R2 = 0.2166
R3 = 0.2166
X = 1.717
R = 1.903
N1 = 0.6637
N2 = 1.302
N3 = 1.302

```

1

```

NUMERO DE USUARIOS= 18
- MEAN VALUE ANALYSIS ("MVA") -
*****
* NAME * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
* * * * *
*CPU *0.3000E-01*0.4316 *0.7180 *0.4991E-01* 14.39 *
* * * * *
*DISC 1 *0.1000 *0.6293 * 1.449 *0.2302 * 6.293 *
* * * * *
*DISC 2 *0.1000 *0.6293 * 1.449 *0.2302 * 6.293 *
* * * * *
*TERMINAL * 8.000 *0.0000E+00* 14.39 * 8.000 * 1.798 *
* * * * *
*****
MEMORY USED: 7825 WORDS OF 4 BYTES
( 0.16 % OF TOTAL MEMORY)
R1 = 0.4991E-01
R2 = 0.2302
R3 = 0.2302
X = 1.798
R = 2.010
N1 = 0.7180
N2 = 1.449
N3 = 1.449

```

1

```

NUMERO DE USUARIOS= 19
- MEAN VALUE ANALYSIS ("MVA") -
*****
* NAME * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
* * * * *
*CPU *0.3000E-01*0.4503 *0.7736 *0.5154E-01* 15.01 *
* * * * *
*DISC 1 *0.1000 *0.6567 * 1.608 *0.2449 * 6.567 *
* * * * *
*DISC 2 *0.1000 *0.6567 * 1.608 *0.2449 * 6.567 *
* * * * *
*TERMINAL * 8.000 *0.0000E+00* 15.01 * 8.000 * 1.876 *
* * * * *
*****
MEMORY USED: 7848 WORDS OF 4 BYTES

```

```

      ( 0.16 % OF TOTAL MEMORY)
R1 = 0.5154E-01
R2 = 0.2449
R3 = 0.2449
X = 1.876
R = 2.126
N1 = 0.7736
N2 = 1.608
N3 = 1.608

1
NUMERO DE USUARIOS= 20
- MEAN VALUE ANALYSIS ("MVA") -
*****
* NAME * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
* * * * *
*CPU *0.3000E-01*0.4682 *0.8305 *0.5321E-01* 15.61 *
* * * * *
*DISC 1 *0.1000 *0.6828 * 1.781 *0.2608 * 6.828 *
* * * * *
*DISC 2 *0.1000 *0.6828 * 1.781 *0.2608 * 6.828 *
* * * * *
*TERMINAL * 8.000 *0.0000E+00* 15.61 * 8.000 * 1.951 *
* * * * *
*****
MEMORY USED: 7871 WORDS OF 4 BYTES
      ( 0.16 % OF TOTAL MEMORY)
R1 = 0.5321E-01
R2 = 0.2608
R3 = 0.2608
X = 1.951
R = 2.251
N1 = 0.8305
N2 = 1.781
N3 = 1.781

```

```

1
NUMERO DE USUARIOS= 21
- MEAN VALUE ANALYSIS ("MVA") -
*****
* NAME * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
* * * * *
*CPU *0.3000E-01*0.4853 *0.8883 *0.5491E-01* 16.18 *
* * * * *
*DISC 1 *0.1000 *0.7077 * 1.968 *0.2781 * 7.077 *
* * * * *
*DISC 2 *0.1000 *0.7077 * 1.968 *0.2781 * 7.077 *
* * * * *
*TERMINAL * 8.000 *0.0000E+00* 16.18 * 8.000 * 2.022 *
* * * * *
*****
MEMORY USED: 7894 WORDS OF 4 BYTES
      ( 0.16 % OF TOTAL MEMORY)
R1 = 0.5491E-01
R2 = 0.2781
R3 = 0.2781

```

X = 2.022
 R = 2.386
 N1 = 0.8883
 N2 = 1.968
 N3 = 1.968

1

NUMERO DE USUARIOS= 22
 - MEAN VALUE ANALYSIS ("MVA") -

```
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *         *
*CPU       *0.3000E-01*0.5014  *0.9468  *0.5665E-01* 16.71  *
*          *         *         *         *         *         *
*DISC  1   *0.1000    *0.7312  * 2.170  *0.2968    * 7.312  *
*          *         *         *         *         *         *
*DISC  2   *0.1000    *0.7312  * 2.170  *0.2968    * 7.312  *
*          *         *         *         *         *         *
*TERMINAL * 8.000     *0.0000E+00* 16.71  * 8.000    * 2.089  *
*          *         *         *         *         *         *
*****
      MEMORY USED:      7917 WORDS OF 4 BYTES
      ( 0.16 % OF TOTAL MEMORY)
```

R1 = 0.5665E-01
 R2 = 0.2968
 R3 = 0.2968
 X = 2.089
 R = 2.531
 N1 = 0.9468
 N2 = 2.170
 N3 = 2.170

1

NUMERO DE USUARIOS= 23
 - MEAN VALUE ANALYSIS ("MVA") -

```
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *         *
*CPU       *0.3000E-01*0.5165  * 1.006  *0.5840E-01* 17.22  *
*          *         *         *         *         *         *
*DISC  1   *0.1000    *0.7533  * 2.388  *0.3170    * 7.533  *
*          *         *         *         *         *         *
*DISC  2   *0.1000    *0.7533  * 2.388  *0.3170    * 7.533  *
*          *         *         *         *         *         *
*TERMINAL * 8.000     *0.0000E+00* 17.22  * 8.000    * 2.152  *
*          *         *         *         *         *         *
*****
      MEMORY USED:      7940 WORDS OF 4 BYTES
      ( 0.16 % OF TOTAL MEMORY)
```

R1 = 0.5840E-01
 R2 = 0.3170
 R3 = 0.3170
 X = 2.152
 R = 2.686
 N1 = 1.006
 N2 = 2.388

N3 = 2.388

1

```

NUMERO DE USUARIOS=      24
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *         *
*CPU       *0.3000E-01*0.5307  * 1.064  *0.6017E-01* 17.69  *
*          *         *         *         *         *         *
*DISC  1   *0.1000  *0.7740  * 2.622  *0.3388  * 7.740  *
*          *         *         *         *         *         *
*DISC  2   *0.1000  *0.7740  * 2.622  *0.3388  * 7.740  *
*          *         *         *         *         *         *
*TERMINAL * 8.000   *0.0000E+00* 17.69  * 8.000   * 2.211  *
*          *         *         *         *         *         *
*****
          MEMORY USED:      7963 WORDS OF 4 BYTES
          ( 0.16 % OF TOTAL MEMORY)

R1 = 0.6017E-01
R2 = 0.3388
R3 = 0.3388
X = 2.211
R = 2.853
N1 = 1.064
N2 = 2.622
N3 = 2.622

```

1

```

NUMERO DE USUARIOS=      25
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *         *
*CPU       *0.3000E-01*0.5439  * 1.123  *0.6193E-01* 18.13  *
*          *         *         *         *         *         *
*DISC  1   *0.1000  *0.7932  * 2.873  *0.3622  * 7.932  *
*          *         *         *         *         *         *
*DISC  2   *0.1000  *0.7932  * 2.873  *0.3622  * 7.932  *
*          *         *         *         *         *         *
*TERMINAL * 8.000   *0.0000E+00* 18.13  * 8.000   * 2.266  *
*          *         *         *         *         *         *
*****
          MEMORY USED:      7986 WORDS OF 4 BYTES
          ( 0.16 % OF TOTAL MEMORY)

R1 = 0.6193E-01
R2 = 0.3622
R3 = 0.3622
X = 2.266
R = 3.031
N1 = 1.123
N2 = 2.873
N3 = 2.873

```

1

```

NUMERO DE USUARIOS=      26

```

- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*   NAME   * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*           *         *         *         *         *         *
*CPU        *0.3000E-01*0.5561   * 1.181   *0.6369E-01* 18.54 *
*           *         *         *         *         *         *
*DISC  1    *0.1000   *0.8110   * 3.141   *0.3873   * 8.110 *
*           *         *         *         *         *         *
*DISC  2    *0.1000   *0.8110   * 3.141   *0.3873   * 8.110 *
*           *         *         *         *         *         *
*TERMINAL * 8.000    *0.0000E+00* 18.54   * 8.000   * 2.317 *
*           *         *         *         *         *         *
```

MEMORY USED: 8009 WORDS OF 4 BYTES
(0.16 % OF TOTAL MEMORY)

R1 = 0.6369E-01
R2 = 0.3873
R3 = 0.3873
X = 2.317
R = 3.221
N1 = 1.181
N2 = 3.141
N3 = 3.141

1

NUMERO DE USUARIOS= 27

- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*   NAME   * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*           *         *         *         *         *         *
*CPU        *0.3000E-01*0.5673   * 1.237   *0.6542E-01* 18.91 *
*           *         *         *         *         *         *
*DISC  1    *0.1000   *0.8273   * 3.426   *0.4141   * 8.273 *
*           *         *         *         *         *         *
*DISC  2    *0.1000   *0.8273   * 3.426   *0.4141   * 8.273 *
*           *         *         *         *         *         *
*TERMINAL * 8.000    *0.0000E+00* 18.91   * 8.000   * 2.364 *
*           *         *         *         *         *         *
```

MEMORY USED: 8032 WORDS OF 4 BYTES
(0.16 % OF TOTAL MEMORY)

R1 = 0.6542E-01
R2 = 0.4141
R3 = 0.4141
X = 2.364
R = 3.422
N1 = 1.237
N2 = 3.426
N3 = 3.426

1

NUMERO DE USUARIOS= 28

- MEAN VALUE ANALYSIS ("MVA") -

```
*****
*   NAME   * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
```



```

*           *           *           *           *           *
*CPU        *0.3000E-01*0.5776    * 1.292    *0.6711E-01* 19.25    *
*           *           *           *           *           *
*DISC  1    *0.1000    *0.8423    * 3.728    *0.4426    * 8.423    *
*           *           *           *           *           *
*DISC  2    *0.1000    *0.8423    * 3.728    *0.4426    * 8.423    *
*           *           *           *           *           *
*TERMINAL  * 8.000    *0.0000E+00* 19.25    * 8.000    * 2.406    *
*           *           *           *           *           *

```

```
*****
```

MEMORY USED: 8055 WORDS OF 4 BYTES

(0.16 % OF TOTAL MEMORY)

```

R1 = 0.6711E-01
R2 = 0.4426
R3 = 0.4426
X = 2.406
R = 3.635
N1 = 1.292
N2 = 3.728
N3 = 3.728

```

1

NUMERO DE USUARIOS= 29
- MEAN VALUE ANALYSIS ("MVA") -

```
*****
```

```

*   NAME   * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*           *           *           *           *           *
*CPU        *0.3000E-01*0.5869    * 1.345    *0.6876E-01* 19.56    *
*           *           *           *           *           *
*DISC  1    *0.1000    *0.8558    * 4.046    *0.4728    * 8.558    *
*           *           *           *           *           *
*DISC  2    *0.1000    *0.8558    * 4.046    *0.4728    * 8.558    *
*           *           *           *           *           *
*TERMINAL  * 8.000    *0.0000E+00* 19.56    * 8.000    * 2.445    *
*           *           *           *           *           *

```

```
*****
```

MEMORY USED: 8078 WORDS OF 4 BYTES

(0.16 % OF TOTAL MEMORY)

```

R1 = 0.6876E-01
R2 = 0.4728
R3 = 0.4728
X = 2.445
R = 3.860
N1 = 1.345
N2 = 4.046
N3 = 4.046

```

1

NUMERO DE USUARIOS= 30
- MEAN VALUE ANALYSIS ("MVA") -

```
*****
```

```

*   NAME   * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*           *           *           *           *           *
*CPU        *0.3000E-01*0.5953    * 1.396    *0.7035E-01* 19.84    *
*           *           *           *           *           *
*DISC  1    *0.1000    *0.8681    * 4.381    *0.5046    * 8.681    *

```

```

*           *           *           *           *           *           *
*DISC  2  *0.1000    *0.8681    * 4.381    *0.5046    * 8.681    *
*           *           *           *           *           *           *
*TERMINAL * 8.000    *0.0000E+00* 19.84    * 8.000    * 2.480    *
*           *           *           *           *           *           *
*****

```

MEMORY USED: 8101 WORDS OF 4 BYTES

(0.16 % OF TOTAL MEMORY)

```

R1 = 0.7035E-01
R2 = 0.5046
R3 = 0.5046
X = 2.480
R = 4.095
N1 = 1.396
N2 = 4.381
N3 = 4.381

```

1

NUMERO DE USUARIOS= 31
- MEAN VALUE ANALYSIS ("MVA") -

```

*****
*  NAME  *  SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*           *           *           *           *           *           *
*CPU      *0.3000E-01*0.6028    * 1.444    *0.7188E-01* 20.09    *
*           *           *           *           *           *           *
*DISC  1  *0.1000    *0.8791    * 4.730    *0.5381    * 8.791    *
*           *           *           *           *           *           *
*DISC  2  *0.1000    *0.8791    * 4.730    *0.5381    * 8.791    *
*           *           *           *           *           *           *
*TERMINAL * 8.000    *0.0000E+00* 20.09    * 8.000    * 2.512    *
*           *           *           *           *           *           *
*****

```

MEMORY USED: 8124 WORDS OF 4 BYTES

(0.16 % OF TOTAL MEMORY)

```

R1 = 0.7188E-01
R2 = 0.5381
R3 = 0.5381
X = 2.512
R = 4.342
N1 = 1.444
N2 = 4.730
N3 = 4.730

```

1

NUMERO DE USUARIOS= 32
- MEAN VALUE ANALYSIS ("MVA") -

```

*****
*  NAME  *  SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*           *           *           *           *           *           *
*CPU      *0.3000E-01*0.6096    * 1.490    *0.7333E-01* 20.32    *
*           *           *           *           *           *           *
*DISC  1  *0.1000    *0.8890    * 5.095    *0.5730    * 8.890    *
*           *           *           *           *           *           *
*DISC  2  *0.1000    *0.8890    * 5.095    *0.5730    * 8.890    *
*           *           *           *           *           *           *
*TERMINAL * 8.000    *0.0000E+00* 20.32    * 8.000    * 2.540    *
*           *           *           *           *           *           *
*****

```

```

*           *           *           *           *           *           *
*****
      MEMORY USED:      8147 WORDS OF 4 BYTES
      ( 0.16 % OF TOTAL MEMORY)
R1 = 0.7333E-01
R2 = 0.5730
R3 = 0.5730
X = 2.540
R = 4.598
N1 = 1.490
N2 = 5.095
N3 = 5.095

```

```

1
NUMERO DE USUARIOS=      33
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME  *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*           *           *           *           *           *           *
*CPU      *0.3000E-01*0.6157   * 1.533   *0.7470E-01* 20.52   *
*           *           *           *           *           *           *
*DISC  1  *0.1000   *0.8979   * 5.472   *0.6095   * 8.979   *
*           *           *           *           *           *           *
*DISC  2  *0.1000   *0.8979   * 5.472   *0.6095   * 8.979   *
*           *           *           *           *           *           *
*TERMINAL * 8.000    *0.0000E+00* 20.52   * 8.000    * 2.565   *
*           *           *           *           *           *           *
*****
      MEMORY USED:      8170 WORDS OF 4 BYTES
      ( 0.16 % OF TOTAL MEMORY)
R1 = 0.7470E-01
R2 = 0.6095
R3 = 0.6095
X = 2.565
R = 4.864
N1 = 1.533
N2 = 5.472
N3 = 5.472

```

```

1
NUMERO DE USUARIOS=      34
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME  *  SERVICE *  BUSY PCT *  CUST NB *  RESPONSE *  THRUPUT *
*****
*           *           *           *           *           *           *
*CPU      *0.3000E-01*0.6211   * 1.573   *0.7599E-01* 20.70   *
*           *           *           *           *           *           *
*DISC  1  *0.1000   *0.9057   * 5.862   *0.6472   * 9.057   *
*           *           *           *           *           *           *
*DISC  2  *0.1000   *0.9057   * 5.862   *0.6472   * 9.057   *
*           *           *           *           *           *           *
*TERMINAL * 8.000    *0.0000E+00* 20.70   * 8.000    * 2.588   *
*           *           *           *           *           *           *
*****
      MEMORY USED:      8193 WORDS OF 4 BYTES
      ( 0.16 % OF TOTAL MEMORY)

```

R1 = 0.7599E-01
 R2 = 0.6472
 R3 = 0.6472
 X = 2.588
 R = 5.138
 N1 = 1.573
 N2 = 5.862
 N3 = 5.862

1

NUMERO DE USUARIOS= 35
 - MEAN VALUE ANALYSIS ("MVA") -

* NAME	* SERVICE	* BUSY PCT	* CUST NB	* RESPONSE	* THRUPUT
*CPU	*0.3000E-01	*0.6259	* 1.611	*0.7720E-01	* 20.86
*DISC 1	*0.1000	*0.9127	* 6.263	*0.6862	* 9.127
*DISC 2	*0.1000	*0.9127	* 6.263	*0.6862	* 9.127
*TERMINAL	* 8.000	*0.0000E+00	* 20.86	* 8.000	* 2.608

 MEMORY USED: 8216 WORDS OF 4 BYTES
 (0.16 % OF TOTAL MEMORY)

R1 = 0.7720E-01
 R2 = 0.6862
 R3 = 0.6862
 X = 2.608
 R = 5.421
 N1 = 1.611
 N2 = 6.263
 N3 = 6.263

1

NUMERO DE USUARIOS= 36
 - MEAN VALUE ANALYSIS ("MVA") -

* NAME	* SERVICE	* BUSY PCT	* CUST NB	* RESPONSE	* THRUPUT
*CPU	*0.3000E-01	*0.6302	* 1.645	*0.7832E-01	* 21.01
*DISC 1	*0.1000	*0.9190	* 6.675	*0.7263	* 9.190
*DISC 2	*0.1000	*0.9190	* 6.675	*0.7263	* 9.190
*TERMINAL	* 8.000	*0.0000E+00	* 21.01	* 8.000	* 2.626

 MEMORY USED: 8239 WORDS OF 4 BYTES
 (0.16 % OF TOTAL MEMORY)

R1 = 0.7832E-01
 R2 = 0.7263
 R3 = 0.7263
 X = 2.626

R = 5.711
 N1 = 1.645
 N2 = 6.675
 N3 = 6.675

1

```

NUMERO DE USUARIOS=      37
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.6340  * 1.677  *0.7935E-01* 21.13 *
*          *          *          *          *          *
*DISC  1   *0.1000  *0.9245  * 7.096  *0.7675  * 9.245  *
*          *          *          *          *          *
*DISC  2   *0.1000  *0.9245  * 7.096  *0.7675  * 9.245  *
*          *          *          *          *          *
*TERMINAL * 8.000   *0.0000E+00* 21.13  * 8.000   * 2.641  *
*          *          *          *          *          *
*****
      MEMORY USED:      8262 WORDS OF 4 BYTES
      ( 0.17 % OF TOTAL MEMORY)

R1 = 0.7935E-01
R2 = 0.7675
R3 = 0.7675
X = 2.641
R = 6.007
N1 = 1.677
N2 = 7.096
N3 = 7.096

```

1

```

NUMERO DE USUARIOS=      38
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.6373  * 1.706  *0.8031E-01* 21.24 *
*          *          *          *          *          *
*DISC  1   *0.1000  *0.9295  * 7.525  *0.8096  * 9.295  *
*          *          *          *          *          *
*DISC  2   *0.1000  *0.9295  * 7.525  *0.8096  * 9.295  *
*          *          *          *          *          *
*TERMINAL * 8.000   *0.0000E+00* 21.24  * 8.000   * 2.656  *
*          *          *          *          *          *
*****
      MEMORY USED:      8285 WORDS OF 4 BYTES
      ( 0.17 % OF TOTAL MEMORY)

R1 = 0.8031E-01
R2 = 0.8096
R3 = 0.8096
X = 2.656
R = 6.309
N1 = 1.706
N2 = 7.525
N3 = 7.525

```

1

```

NUMERO DE USUARIOS=      39
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *         *
*CPU       *0.3000E-01*0.6404  * 1.733  *0.8118E-01* 21.35  *
*          *         *         *         *         *         *
*DISC  1   *0.1000    *0.9339  * 7.961  *0.8525   * 9.339  *
*          *         *         *         *         *         *
*DISC  2   *0.1000    *0.9339  * 7.961  *0.8525   * 9.339  *
*          *         *         *         *         *         *
*TERMINAL * 8.000     *0.0000E+00* 21.35  * 8.000   * 2.668  *
*          *         *         *         *         *         *
*****
          MEMORY USED:      8308 WORDS OF 4 BYTES
          ( 0.17 % OF TOTAL MEMORY)

R1 = 0.8118E-01
R2 = 0.8525
R3 = 0.8525
X = 2.668
R = 6.617
N1 = 1.733
N2 = 7.961
N3 = 7.961

```

1

```

NUMERO DE USUARIOS=      40
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *         *         *         *         *         *
*CPU       *0.3000E-01*0.6431  * 1.757  *0.8199E-01* 21.44  *
*          *         *         *         *         *         *
*DISC  1   *0.1000    *0.9378  * 8.403  *0.8961   * 9.378  *
*          *         *         *         *         *         *
*DISC  2   *0.1000    *0.9378  * 8.403  *0.8961   * 9.378  *
*          *         *         *         *         *         *
*TERMINAL * 8.000     *0.0000E+00* 21.44  * 8.000   * 2.679  *
*          *         *         *         *         *         *
*****
          MEMORY USED:      8331 WORDS OF 4 BYTES
          ( 0.17 % OF TOTAL MEMORY)

R1 = 0.8199E-01
R2 = 0.8961
R3 = 0.8961
X = 2.679
R = 6.928
N1 = 1.757
N2 = 8.403
N3 = 8.403

```

1

```

NUMERO DE USUARIOS=      41
- MEAN VALUE ANALYSIS ("MVA") -

```

```
*****
*   NAME   * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*         *         *         *         *         *         *
*CPU      *0.3000E-01*0.6455  * 1.780  *0.8272E-01* 21.52 *
*         *         *         *         *         *         *
*DISC  1  *0.1000  *0.9413  * 8.852  *0.9403  * 9.413  *
*         *         *         *         *         *         *
*DISC  2  *0.1000  *0.9413  * 8.852  *0.9403  * 9.413  *
*         *         *         *         *         *         *
*TERMINAL * 8.000  *0.0000E+00* 21.52  * 8.000  * 2.690  *
*         *         *         *         *         *         *
```

```
*****
MEMORY USED:      8354 WORDS OF 4 BYTES
( 0.17 % OF TOTAL MEMORY)
```

```
R1 = 0.8272E-01
R2 = 0.9403
R3 = 0.9403
X = 2.690
R = 7.244
N1 = 1.780
N2 = 8.852
N3 = 8.852
```

1

```
NUMERO DE USUARIOS=      42
- MEAN VALUE ANALYSIS ("MVA") -
```

```
*****
*   NAME   * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*         *         *         *         *         *         *
*CPU      *0.3000E-01*0.6477  * 1.800  *0.8340E-01* 21.59 *
*         *         *         *         *         *         *
*DISC  1  *0.1000  *0.9445  * 9.305  *0.9852  * 9.445  *
*         *         *         *         *         *         *
*DISC  2  *0.1000  *0.9445  * 9.305  *0.9852  * 9.445  *
*         *         *         *         *         *         *
*TERMINAL * 8.000  *0.0000E+00* 21.59  * 8.000  * 2.699  *
*         *         *         *         *         *         *
```

```
*****
MEMORY USED:      8377 WORDS OF 4 BYTES
( 0.17 % OF TOTAL MEMORY)
```

```
R1 = 0.8340E-01
R2 = 0.9852
R3 = 0.9852
X = 2.699
R = 7.563
N1 = 1.800
N2 = 9.305
N3 = 9.305
```

1

```
NUMERO DE USUARIOS=      43
- MEAN VALUE ANALYSIS ("MVA") -
```

```
*****
*   NAME   * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*         *         *         *         *         *         *
```

```

*CPU      *0.3000E-01*0.6496      * 1.819      *0.8401E-01* 21.65      *
*          *          *          *          *          *          *
*DISC  1  *0.1000      *0.9474      * 9.763      * 1.031      * 9.474      *
*          *          *          *          *          *          *
*DISC  2  *0.1000      *0.9474      * 9.763      * 1.031      * 9.474      *
*          *          *          *          *          *          *
*TERMINAL * 8.000      *0.0000E+00* 21.65      * 8.000      * 2.707      *
*          *          *          *          *          *          *

```

```

*****

```

MEMORY USED: 8400 WORDS OF 4 BYTES

(0.17 % OF TOTAL MEMORY)

R1 = 0.8401E-01

R2 = 1.031

R3 = 1.031

X = 2.707

R = 7.886

N1 = 1.819

N2 = 9.763

N3 = 9.763

1

NUMERO DE USUARIOS= 44

- MEAN VALUE ANALYSIS ("MVA") -

```

*****

```

```

*  NAME      * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *          *
*CPU      *0.3000E-01*0.6514      * 1.837      *0.8458E-01* 21.71      *
*          *          *          *          *          *          *
*DISC  1  *0.1000      *0.9500      * 10.22      * 1.076      * 9.500      *
*          *          *          *          *          *          *
*DISC  2  *0.1000      *0.9500      * 10.22      * 1.076      * 9.500      *
*          *          *          *          *          *          *
*TERMINAL * 8.000      *0.0000E+00* 21.71      * 8.000      * 2.714      *
*          *          *          *          *          *          *

```

```

*****

```

MEMORY USED: 8423 WORDS OF 4 BYTES

(0.17 % OF TOTAL MEMORY)

R1 = 0.8458E-01

R2 = 1.076

R3 = 1.076

X = 2.714

R = 8.211

N1 = 1.837

N2 = 10.22

N3 = 10.22

1

NUMERO DE USUARIOS= 45

- MEAN VALUE ANALYSIS ("MVA") -

```

*****

```

```

*  NAME      * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*          *          *          *          *          *          *
*CPU      *0.3000E-01*0.6530      * 1.852      *0.8510E-01* 21.77      *
*          *          *          *          *          *          *
*DISC  1  *0.1000      *0.9523      * 10.69      * 1.122      * 9.523      *
*          *          *          *          *          *          *

```



```
*DISC  2  *0.1000    *0.9523    * 10.69    * 1.122    * 9.523    *
*
*
*TERMINAL * 8.000    *0.0000E+00* 21.77    * 8.000    * 2.721    *
*
*****
```

```
MEMORY USED:      8446 WORDS OF 4 BYTES
( 0.17 % OF TOTAL MEMORY)
```

```
R1 = 0.8510E-01
R2 = 1.122
R3 = 1.122
X = 2.721
R = 8.538
N1 = 1.852
N2 = 10.69
N3 = 10.69
```

1

```
NUMERO DE USUARIOS=      46
- MEAN VALUE ANALYSIS ("MVA") -
```

```
*****
*  NAME    * SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *         *         *          *         *
*CPU       *0.3000E-01*0.6545  * 1.867   *0.8557E-01* 21.82  *
*          *         *         *          *         *
*DISC  1   *0.1000    *0.9545  * 11.16   * 1.169    * 9.545  *
*          *         *         *          *         *
*DISC  2   *0.1000    *0.9545  * 11.16   * 1.169    * 9.545  *
*          *         *         *          *         *
*TERMINAL * 8.000    *0.0000E+00* 21.82   * 8.000    * 2.727  *
*          *         *         *          *         *
```

```
MEMORY USED:      8469 WORDS OF 4 BYTES
( 0.17 % OF TOTAL MEMORY)
```

```
R1 = 0.8557E-01
R2 = 1.169
R3 = 1.169
X = 2.727
R = 8.867
N1 = 1.867
N2 = 11.16
N3 = 11.16
```

1

```
NUMERO DE USUARIOS=      47
- MEAN VALUE ANALYSIS ("MVA") -
```

```
*****
*  NAME    * SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *         *         *          *         *
*CPU       *0.3000E-01*0.6559  * 1.880   *0.8601E-01* 21.86  *
*          *         *         *          *         *
*DISC  1   *0.1000    *0.9565  * 11.63   * 1.216    * 9.565  *
*          *         *         *          *         *
*DISC  2   *0.1000    *0.9565  * 11.63   * 1.216    * 9.565  *
*          *         *         *          *         *
*TERMINAL * 8.000    *0.0000E+00* 21.86   * 8.000    * 2.733  *
*          *         *         *          *         *
```

```
*****
MEMORY USED:      8492 WORDS OF 4 BYTES
( 0.17 % OF TOTAL MEMORY)
R1 = 0.8601E-01
R2 = 1.216
R3 = 1.216
X = 2.733
R = 9.199
N1 = 1.880
N2 = 11.63
N3 = 11.63
```

1

```
NUMERO DE USUARIOS=      48
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME  * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*      *      *      *      *      *      *
*CPU    *0.3000E-01*0.6571 * 1.893 *0.8641E-01* 21.90 *
*      *      *      *      *      *      *
*DISC  1 *0.1000 *0.9583 * 12.10 * 1.263 * 9.583 *
*      *      *      *      *      *      *
*DISC  2 *0.1000 *0.9583 * 12.10 * 1.263 * 9.583 *
*      *      *      *      *      *      *
*TERMINAL * 8.000 *0.0000E+00* 21.90 * 8.000 * 2.738 *
*      *      *      *      *      *      *
```

```
*****
MEMORY USED:      8515 WORDS OF 4 BYTES
( 0.17 % OF TOTAL MEMORY)
R1 = 0.8641E-01
R2 = 1.263
R3 = 1.263
X = 2.738
R = 9.531
N1 = 1.893
N2 = 12.10
N3 = 12.10
```

1

```
NUMERO DE USUARIOS=      49
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME  * SERVICE * BUSY PCT * CUST NB * RESPONSE * THRUPUT *
*****
*      *      *      *      *      *      *
*CPU    *0.3000E-01*0.6582 * 1.904 *0.8678E-01* 21.94 *
*      *      *      *      *      *      *
*DISC  1 *0.1000 *0.9599 * 12.58 * 1.310 * 9.599 *
*      *      *      *      *      *      *
*DISC  2 *0.1000 *0.9599 * 12.58 * 1.310 * 9.599 *
*      *      *      *      *      *      *
*TERMINAL * 8.000 *0.0000E+00* 21.94 * 8.000 * 2.743 *
*      *      *      *      *      *      *
```

```
*****
MEMORY USED:      8538 WORDS OF 4 BYTES
( 0.17 % OF TOTAL MEMORY)
R1 = 0.8678E-01
```

```

R2 =    1.310
R3 =    1.310
X =    2.743
R =    9.866
N1 =    1.904
N2 =    12.58
N3 =    12.58

```

1

```

NUMERO DE USUARIOS=      50
- MEAN VALUE ANALYSIS ("MVA") -
*****
*  NAME    * SERVICE * BUSY PCT *  CUST NB * RESPONSE *  THRUPUT *
*****
*          *          *          *          *          *
*CPU       *0.3000E-01*0.6593  *  1.915  *0.8712E-01* 21.98  *
*          *          *          *          *          *
*DISC  1   *0.1000    *0.9615  * 13.05   * 1.358   * 9.615   *
*          *          *          *          *          *
*DISC  2   *0.1000    *0.9615  * 13.05   * 1.358   * 9.615   *
*          *          *          *          *          *
*TERMINAL * 8.000     *0.0000E+00* 21.98   * 8.000   * 2.747   *
*          *          *          *          *          *
*****
          MEMORY USED:      8561 WORDS OF 4 BYTES
          ( 0.17  % OF TOTAL MEMORY)
R1 =    0.8712E-01
R2 =    1.358
R3 =    1.358
X =    2.747
R =    10.20
N1 =    1.915
N2 =    13.05
N3 =    13.05
      49 /END/

```